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A SECOND MAIN THEOREM FOR ENTIRE CURVES INTERSECTING THREE CONICS

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ABSTRACT. We establish a Second Main Theorem for algebraically nondegenerate entire holomorphic curves $f:\mathbb{C}\to\mathbb{P}^2$ intersecting a generic configuration of three conics $\mathscr{C}=\mathscr{C}_1+\mathscr{C}_2+\mathscr{C}_3$ in the complex projective plane $\mathbb{P}^2$. Using invariant logarithmic 2-jet differentials with negative twists, we prove the estimate

$$T_f(r)\leqslant 5\sum_{i=1}^{3}N_f^{[1]}(r,\mathscr{C}_i)+o\big(T_f(r)\big)\qquad\|,$$

where $T_f(r)$ is the Nevanlinna characteristic function, and $N_f^{[1]}(r,\mathscr{C}_i)$ is the 1-truncated counting function.

We also prove that the logarithmic cotangent bundle of three generic conics is nef and that all of its negatively twisted symmetric powers have no nonzero global sections. Consequently, within the negatively twisted jet-differential approach, order two is the first level at which an effective argument can operate.

The key innovation of our approach is establishing new vanishing lemmas of the form

$$H^0\big(\mathbb{P}^2,E_{2,m}T^*_{\mathbb{P}^2}(\log\mathscr{C})\otimes\mathscr{O}_{\mathbb{P}^2}(-t)\big)=0$$

for specific pairs (m,t), achieved by combining algebro-geometric arguments with computer-assisted computations through a mod-p reduction technique. This provides a systematic method for proving vanishing results for negatively twisted jet differentials — a key component in complex hyperbolic geometry — thereby filling a missing piece in two-dimensional complex hyperbolicity theory.

1. Introduction

1.1. **Motivation.** In the complex projective plane $\mathbb{P}^2$, the *logarithmic Kobayashi conjecture* asserts that if $\mathscr{C}\subset\mathbb{P}^2$ is a *generic* algebraic curve of degree $d\geqslant 5$, then the complement $\mathbb{P}^2\setminus\mathscr{C}$ should be hyperbolic [39]. Various notions of hyperbolicity exist; here we use the most straightforward one: $\mathbb{P}^2\setminus\mathscr{C}$ is said to be *hyperbolic* if there is no nonconstant holomorphic map $f:\mathbb{C}\to\mathbb{P}^2\setminus\mathscr{C}$. This conjecture generalizes the classical Little Picard Theorem on the hyperbolicity of $\mathbb{P}^1\setminus\{3\text{ points}\}$, and is supported by explicit examples constructed by Zaidenberg [75]. For generic curves $\mathscr{C}$ of sufficiently large degree in $\mathbb{P}^2$, the conjecture was confirmed by Siu and Yeung [67] by constructing explicit logarithmic 2-jet differentials with negative twists. Later, building on the work [18], El Goul [25] improved the degree bound to 15 for *very generic* $\mathscr{C}$. Combining this with Siu's strategy of using slanted vector fields [65] and a technique of Păun [54], Rousseau [58] further reduced the degree bound to 14.

These results for high-degree curves naturally extend to configurations with multiple components. In the case where the curve $\mathscr{C}=\cup_{i=1}^q\mathscr{C}_i$ decomposes into the union of $q\geqslant 2$ generic algebraic curves $\mathscr{C}_i$ of degree d_i, the hyperbolicity of the complement $\mathbb{P}^2\setminus\mathscr{C}$ has been established with smaller degree bounds $d=\sum_{i=1}^q d_i$ in various works. For instance, a result in [1] settles the case where $q\geqslant 5$. The case $(q=4,\sum_{i=1}^4 d_i\geqslant 5)$ was treated in [21]. The configuration of three conics was studied in [29, 21, 2] by different methods. Dethloff, Schumacher, and Wong treated the case of two conics and one cubic using symmetric differential 1-forms [22], and settled the case of three generic conics by a clever application of Nevanlinna theory [21]. Rousseau [58] addressed the case $q=2$ under the assumptions $(d_1=1,d_2\geqslant 11)$, $(d_1=2,d_2\geqslant 8)$, $(d_1=3,d_2\geqslant 5)$, or $(d_1=4,d_2\geqslant 4)$. Notably, for $q\geqslant 3$, Noguchi, Winkelmann, and Yamanoi [53] established the *algebraic degeneracy* of every entire curve in $\mathbb{P}^2\setminus\mathscr{C}$ under the mild condition $\sum_{i=1}^q d_i\geqslant 4$. Their result also extends to the corresponding higher-dimensional setting.

Thus, in the aforementioned settings, every algebraically nondegenerate entire curve $f:\mathbb{C}\to\mathbb{P}^2$ must intersect the curve $\mathscr{C}$. In this article, we aim to quantify these geometric results by means of Nevanlinna

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theory. Specifically, we aim to establish a Second Main Theorem type estimate:

$$T_f(r) \leqslant c\sum_{i=1}^{q} N_f^{[k]}(r,\mathscr{C}_i) + o\big(T_f(r)\big) \quad \|, \tag{1.1}$$

where

$$N_f^{[k]}(r,\mathscr{C}_i) := \int_1^r \sum_{|z|<t} \min\{k, \mathrm{ord}_z f^*\mathscr{C}_i\}\,\frac{\mathrm{d}t}{t} \quad (i=1,\dots,q;\, r>1)$$

are the *k-truncated counting functions*, measuring the frequency of intersections $f(\mathbb{C})\cap\mathscr{C}_i$; and

$$T_f(r) := \int_1^r \frac{\mathrm{d}t}{t}\int_{|z|<t} f^*\omega_{\mathrm{FS}} \quad (r>1)$$

is the *order function*, capturing the area growth rate of the holomorphic discs $f(\{|z|<t\})$ with respect to the Fubini-Study metric ω_{FS} on $\mathbb{P}^2$. The constant c on the right-hand side of (1.1) is independent of r, and the notation "$\|$" indicates that the inequality (1.1) holds for all $r>1$ outside a set of finite Lebesgue measure.

When $\mathscr{C}$ has at least 4 irreducible components, such a Second Main Theorem type estimate with truncation at level 2 was established by H. Cartan [7] in the linear case where $\mathscr{C}_i$ are lines:

$$(q-3)T_f(r) \leqslant \sum_{i=1}^{q} N_f^{[2]}(r,\mathscr{C}_i) + o\big(T_f(r)\big) \quad \| \,.$$

In the nonlinear case, analogous results (without truncation) were obtained by Eremenko and Sodin [27] via pluripotential theory, and by Ru [60] through a synthesis of Cartan's method and Diophantine geometry. These results also extend to the corresponding higher-dimensional setting (see [51, 61]).

The above estimate becomes vacuous when $q=3$, as the factor $(q-3)$ vanishes; thus the case of three curves lies precisely on the borderline where classical methods break down. From the viewpoint of the logarithmic Bloch–Ochiai theorem, the condition $q>3$ was originally required to guarantee algebraic degeneracy of every entire curve $f:\mathbb{C}\to\mathbb{P}^2\setminus\bigcup_{i=1}^q\mathscr{C}_i$ (see the series of works by Noguchi [47, 48, 49]).

In the prior work [10], the authors established a Second Main Theorem for $n+1$ hypersurfaces $\{D_i\}_{i=1}^{n+1}$ in $\mathbb{P}^n$ in general position and intersecting transversally, proving the defect relation

$$\sum_{i=1}^{n+1}\delta_f(D_i) < n+1 \tag{1.2}$$

for any algebraically nondegenerate entire curve $f:\mathbb{C}\to\mathbb{P}^n$ under the optimal total degree assumption $\sum_{i=1}^{n+1}\deg D_i \geqslant n+2$. Here, each defect $\delta_f(D_i)$ is defined by

$$\delta_f(D_i) := 1 - \limsup_{r\to\infty}\frac{N_f(r,D_i)}{\deg(D_i)\,T_f(r)}.$$

The proof combined parabolic Nevanlinna theory [55] with the Second Main Theorems for semi-Abelian varieties [52] and for $n+2$ hypersurfaces [59]. Subsequently, Ru and Wang [62] extended the result (1.2) to moving targets with truncation level 1, while preserving the upper bound $n+1$ on the right-hand side (see also [63]).

In the case of $n=2$, the defect relation (1.2) might be viewed as the weakest, albeit nontrivial, form of a Second Main Theorem in the vein of (1.1) with noneffective $c<\infty$. The primary objective of this article is to establish an effective Second Main Theorem, with truncation level 1, for three generic conics with an explicit constant $c=5$.

1.2. **Two Key Theorems.** The pursuit of a Second Main Theorem encounters a fundamental obstacle: the very nature of entire curves exhibits an inherent and wild flexibility in large-scale geometry, as evidenced by the exotic examples in [38, 73, 71, 9]. Our strategy constrains this flexibility by leveraging additional algebro-geometric structures to extract the underlying rigidity of entire curves, building on the following two key results.

Theorem 1.1 (McQuillan [41])**.** *Let $\mathscr{C}=\sum_{i=1}^q\mathscr{C}_i$ be a normal crossings divisor in $\mathbb{P}^2$, where each $\mathscr{C}_i$ is a smooth algebraic curve of degree d_i with $d=\sum_{i=1}^q d_i\geqslant 4$. Let $f:\mathbb{C}\to\mathbb{P}^2$ be an algebraically nondegenerate entire holomorphic curve. If the lifting $f_{[1]}:\mathbb{C}\to X_1=\mathbb{P}T_{\mathbb{P}^2}(-\log\mathscr{C})$ is algebraically degenerate, then the following Second Main Theorem type estimate holds:*

$$(d-3)\,T_f(r) \leqslant N_f^{[1]}(r,\mathscr{C}) + o\big(T_f(r)\big) \quad \| \,.$$

Theorem 1.2 ([37, Theorem 3.1]). *Let $\mathscr{C} = \sum_{i=1}^{q} \mathscr{C}_i$ be a normal crossings divisor in $\mathbb{P}^2$. For any logarithmic jet differential*

$$\omega \in H^0\big(\mathbb{P}^2, E_{k,m}^{GG} T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big) \quad (k,m,t \geqslant 1),$$

and any entire holomorphic curve $f : \mathbb{C} \to \mathbb{P}^2$ such that $f^*\omega \not\equiv 0$, the following Second Main Theorem type estimate holds:

$$T_f(r) \leqslant \frac{m}{t} \sum_{i=1}^{q} N_f^{[1]}(r, \mathscr{C}_i) + O\big(\log T_f(r) + \log r\big) \quad \| \,. \tag{1.3}$$

Here, $E_{k,m}^{GG} T^*_{\mathbb{P}^2}(\log \mathscr{C})$ denotes the logarithmic Green–Griffiths k-jet bundle of weighted order m (see Section 2). The use of such higher-order jet differentials in the study of complex hyperbolicity goes back to André Bloch [3]. In the present work, we restrict attention to the "invariant" subbundle $E_{k,m} T^*_{\mathbb{P}^2}(\log \mathscr{C}) \subset E_{k,m}^{GG} T^*_{\mathbb{P}^2}(\log \mathscr{C})$, originally introduced by Demailly [17]. This subbundle possesses stronger positivity properties, making it particularly suitable for constructing sections; further details can be found in Section 2.

Theorem 1.2 can be seen as a quantitative refinement of the Fundamental Vanishing Theorem for entire curves (cf. [30, pp. 55-59], [66, Lemma 1.2] and [17, Corollary 7.9]). Although the original statement in [37] is formulated for hypersurfaces in $\mathbb{P}^n$, the proof extends directly to the case of simple normal crossing divisors in projective manifolds (cf. [61, Theorem A10.7.2]).

1.3. **Our First Result.** The first result of this article is an application of Theorem 1.2 in the case $k = 1$.

Theorem 1.3. *Let $\mathscr{C} = \cup_{i=1}^{3} \mathscr{C}_i$ be a union of three smooth curves $\mathscr{C}_i$ of degrees d_i with simple normal crossings. Assuming that $d_1 \geqslant d_2 \geqslant d_3 \geqslant 2$ and $d_1 \geqslant 3$, so that the total degree $d = d_1 + d_2 + d_3 \geqslant 7$. For a nonconstant algebraically nondegenerate entire holomorphic curve $f\colon \mathbb{C} \to \mathbb{P}^2$ whose image is not contained in any $\mathscr{C}_i$, one has the estimate*

$$T_f(r) \leqslant c \sum_{i=1}^{3} N_f^{[1]}(r, \mathscr{C}_i) + o\big(T_f(r)\big) \quad \|,$$

for any positive constant c satisfying

$$c > \frac{1}{(d-3)\delta_1}, \quad \textit{where} \quad \delta_1 = \tfrac{1}{2} - \frac{\sqrt{9(d-1)^2 - 12(d_1d_2 + d_2d_3 + d_3d_1)}}{6(d-3)}.$$

Here, the numerical conditions on d_1, d_2, d_3 are imposed to ensure the positivity of the Euler characteristic of the logarithmic symmetric differential bundles $S^m T^*_{\mathbb{P}^2}(\log \mathscr{C})$. Specifically, we have:

$$\chi\big(S^m T^*_{\mathbb{P}^2}(\log \mathscr{C})\big) = L_{d_1d_2d_3} \cdot m^3 + O(m^2) > 0, \quad \forall m \gg 1,$$

where $L_{d_1d_2d_3}$ is a positive leading coefficient.

Remark 1.4. In the case where $\mathscr{C}$ consists of two conics and one cubic, a direct computation shows that δ_1 takes the value $\frac{6-\sqrt{33}}{12}$. This leads to a Second Main Theorem of the form

$$T_f(r) \leqslant c \sum_{i=1}^{3} N_f^{[1]}(r, \mathscr{C}_i) + o\big(T_f(r)\big) \quad \|,$$

valid for any positive constant c satisfying $c > \frac{3}{6-\sqrt{33}} \approx 11.74\ldots$. This quantifies the hyperbolicity result of $\mathbb{P}^2 \setminus \mathscr{C}$ due to Dethloff–Schumacher–Wong [22].

1.4. **Proof of Theorem 1.3.** Let f be algebraically nondegenerate, as in the statement of the theorem. The key step is to find a nonzero global negatively twisted logarithmic symmetric differential form

$$\omega \in H^0\big(\mathbb{P}^2, S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}(-t)\big), \quad m \geqslant 1,\ t \geqslant 1,\ m/t \leqslant c,\ \omega \not\equiv 0. \tag{1.4}$$

Proof of Theorem 1.3 assuming (1.4). If $f^*\omega \not\equiv 0$, then Theorem 1.2 yields the desired estimate. On the other hand, if $f^*\omega \equiv 0$, then $f_{[1]}(\mathbb{C})$ lies in the proper zero divisor of ω on X_1. We can therefore apply McQuillan's Theorem 1.1 to obtain the stronger inequality

$$(d-3)\, T_f(r) \leqslant N_f^{[1]}(r, \mathscr{C}) + o(T_f(r)) \quad \| \,.$$

In either case, the conclusion of the theorem holds. □

The proof of (1.4) will be given in Section 3, based on Riemann–Roch computations and a Bogomolov-type vanishing theorem (Theorem 2.1).

1.5. **Our Second, Main Result: Second Main Theorem for Three Conics.** Having established an effective Second Main Theorem for configurations of large degree, we now turn to the more challenging case of three generic conics $\mathscr{C} = \mathscr{C}_1 + \mathscr{C}_2 + \mathscr{C}_3$ in $\mathbb{P}^2$. The residue sequence gives $c_1(T^*_{\mathbb{P}^2}(\log \mathscr{C})) = 3h$ and $c_2(T^*_{\mathbb{P}^2}(\log \mathscr{C})) = 9h^2$. Here h represents the class of $\mathscr{O}_{\mathbb{P}^2}(1)$, so satisfies $h^2 = 1$. Consequently, Hirzebruch–Riemann–Roch yields the exact formula

$$\chi\big(S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big) = \frac{m+1}{2}\big((t-1)(t-2) - 3mt\big). \tag{1.5}$$

In particular, $\chi(S^m T^*_{\mathbb{P}^2}(\log \mathscr{C})) = m+1$; thus the untwisted Euler characteristic does not have a negative cubic leading term. This naturally raises the corresponding one-jet vanishing question, which is completely resolved by the following theorem.

Proposition 1.5 (One-jet vanishing for three conics)**.** *For a generic simple-normal-crossing configuration of three smooth conics $\mathscr{C} = \mathscr{C}_1 + \mathscr{C}_2 + \mathscr{C}_3$ in $\mathbb{P}^2$, the logarithmic cotangent bundle*

$$E = T^*_{\mathbb{P}^2}(\log \mathscr{C})$$

is nef. Moreover, for every pair of integers $m, t \geqslant 1$,

$$H^0\big(\mathbb{P}^2, S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big) = 0. \tag{1.6}$$

The proof is given in Appendix B. It is instructive to compare the theorem with known borderline cases. For certain special configurations of six lines, negatively twisted logarithmic symmetric differentials do exist; see [34, p. 133] and [69, Theorem 1.4]. For a generic sextic curve, they do not exist [25, Lemma 1.4.1]. Proposition 1.5 shows that three generic conics satisfy the stronger vanishing for every negative twist. Thus the one-jet route is genuinely unavailable in this low-degree setting. Invariant logarithmic 2-jets are therefore not merely a technical convenience: they are the first possible jet order for the negatively twisted jet-differential strategy. This necessity gives the two-jet construction in the present paper additional geometric significance. This is a precise instance of the principle that "*the lower the degree, the harder the hyperbolicity*" [23, p. 3]. Our main result is as follows.

Theorem 1.6 (Second Main Theorem for Three Conics)**.** *For a generic configuration of three conics $\{\mathscr{C}_i\}_{i=1}^3$ in $\mathbb{P}^2$ and a nonconstant algebraically nondegenerate entire holomorphic curve $f\colon \mathbb{C} \to \mathbb{P}^2$ whose image is not contained in any $\mathscr{C}_i$, the following inequality holds:*

$$T_f(r) \leqslant 5 \sum_{i=1}^{3} N_f^{[1]}(r, \mathscr{C}_i) + o\big(T_f(r)\big) \quad \| \,. \tag{1.7}$$

Remark 1.7. The constant $c = 5$ in this theorem is smaller than the constant $\approx$11.74 appearing in Theorem 1.3 for the case of two conics and one cubic. This may seem counterintuitive, since the smaller total degree $2+2+2 < 2+2+3$ would normally suggest a larger constant. Proposition 1.5 shows that negatively twisted 1-jets do not exist for three generic conics; the improvement therefore comes from the more refined invariant 2-jet method. A detailed 2-jet analysis is likewise expected to sharpen the constants for the two-conics-and-one-cubic case and for the other configurations covered by Theorem 1.3.

Remark 1.8. An unexpected computational byproduct appears for the three Fermat-type conics $\mathscr{C}$ defined by the quadratic polynomials

$$2Z_0^2 + Z_1^2 + Z_2^2, \qquad Z_0^2 + 2Z_1^2 + Z_2^2, \qquad Z_0^2 + Z_1^2 + 2Z_2^2.$$

For this explicit configuration, the finite calculation detects a 9-dimensional space of negatively twisted invariant logarithmic 2-jets of weighted degree 5 and twist -1; see Remark 7.5. Together with the preceding vanishings in weighted degrees $1, 2, 3, 4$, this forces the space to contain two algebraically independent members and hence yields a Second Main Theorem with constant 5 for this Fermat-type configuration. This byproduct is not used in the proof of the Main Theorem, whose constant is instead governed by the finite set of vanishings in Key Vanishing Lemma II.

Remark 1.9. For a generic configuration of three conics, an effective Second Main Theorem can also be obtained without the algebraic nondegeneracy assumption by combining the results of Song-Yan Xie and Shengyuan Zhao [74] on Poincaré-type degree bounds for invariant algebraic curves of foliations arising from jet differentials with a more refined treatment of algebraically degenerate entire curves. The constant in the estimate without the algebraic nondegeneracy assumption, however, may be larger than 5.

We now outline the main ideas underlying the proof of Theorem 1.6 for a finite constant $c < \infty$, before refining the argument to achieve the sharper goal $c = 5$. Let f be algebraically nondegenerate, as in Theorem 1.6.

The core idea is to construct two independent negatively twisted invariant logarithmic 2-jet differentials ω_1 and ω_2:

- If $f^*\omega_1 \not\equiv 0$ or $f^*\omega_2 \not\equiv 0$, then Theorem 1.2 directly gives the desired Second Main Theorem type estimate.
- If both pullbacks vanish, then their independence forces the lifting $f_{[1]}(\mathbb{C}) \subset X_1$ to be algebraically degenerate. Hence McQuillan's Theorem 1.1 applies.

In either case, we obtain the desired Second Main Theorem type estimate.

In the numerical range where differentiation preserves a negative twist, Siu's slanted vector fields realize this idea directly: the natural action of a slanted vector field on the universal logarithmic jet space sends ω_1 to a genuine global invariant logarithmic 2-jet differential ω_2. When this direct construction is not numerically available, we retain Miyaoka's zero-locus argument (see also Lu–Yau), in the jet-differential form used by Demailly–El Goul [45, 40, 18], as a logically necessary fallback. The conceptual core of our method is the use of two independent negatively twisted invariant logarithmic 2-jet differentials. The zero-locus argument is only a fallback for the remaining numerical cases; sufficiently many additional Key Vanishing Lemmas would rule out those cases and thereby make the fallback unnecessary.

The central difficulty is to control the common base locus of the resulting invariant logarithmic 2-jet differentials. Constructions with sufficiently controlled base locus are available in high degree, as in Siu–Yeung [67], but become far more difficult in low degree because of the algebraic complexity of 2-jet differentials and the associated elimination problem; compare [42]. In the case of three generic conics, this difficulty cannot be bypassed by reverting to the simpler first-order argument: Proposition 1.5 shows that no negatively twisted logarithmic 1-jet differentials exist.

A notable illustration of these difficulties arises even in the 1-jet case. Debarre [16] conjectured that for a generic choice of high-degree hypersurfaces $H_1, \dots, H_c \subset \mathbb{P}^N$ with $c \geqslant N/2$, the complete intersection $X := H_1 \cap \cdots \cap H_c$ should have an ample cotangent bundle T^*_X, which is equivalent to the existence of sufficiently many negatively twisted 1-jet differentials on X with empty common base locus. The first proof of Debarre's ampleness conjecture was established by Xie [72], using explicit 1-jet differentials obtained by Brotbek [5] for Fermat-type hypersurfaces. A key step in Xie's approach is a core lemma [72, p. 969] that overcomes substantial difficulties in the elimination process. Shortly thereafter, Brotbek and Darondeau [6] found an alternative proof, which instead applies Nakamaye's theorem [46] to tackle the elimination.

1.6. **Roadmap for Theorem 1.6: Towards an Effective Finite Constant $c < \infty$.** We now outline the proof of our main result, focusing first on establishing an explicit finite constant c in the inequality (1.7). The argument proceeds in several steps.

First, for generic three conics $\mathscr{C} = \mathscr{C}_1 + \mathscr{C}_2 + \mathscr{C}_3$, standard Riemann–Roch computations (detailed in Section 4) guarantee the existence of an irreducible, negatively twisted invariant logarithmic 2-jet differential

$$\omega_1 \in H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big), \tag{1.8}$$

with weighted degree $m \geqslant 1$, vanishing order $t \geqslant 1$.

If $f^*\omega_1 \not\equiv 0$, then Theorem 1.2 immediately yields the desired Second Main Theorem with a finite constant c. We may therefore assume $f^*\omega_1 \equiv 0$ in what follows.

The subsequent strategy depends on the values of (m,t):

Case 1: $t > 7$. We apply Siu's strategy [65] of slanted vector fields, as realized in dimension two by Păun [54] (compact case) and Rousseau [58] (logarithmic case). On the universal logarithmic jet space, a pole-order-7 slanted vector field acts naturally on the weighted homogeneous polynomial associated with a holomorphic family extending ω_1 and produces a genuine global invariant logarithmic 2-jet differential

$$\omega_2 \in H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(7-t)\big)\,.$$

The field can be chosen so that ω_1 and ω_2 are independent modulo Γ_2. Theorem 1.2 applied to ω_2 gives the coefficient $m/(t-7)$; if both pullbacks vanish, their independence forces $f_{[1]}$ to be algebraically degenerate. See Section 5.

Case 2: $t < m$**.** Following Miyaoka's zero-locus method (see also Lu–Yau), in the form developed by Demailly–El Goul [45, 40, 18] (see Section 6), Riemann–Roch computations give a nonzero negatively twisted section on the residual zero divisor of ω_1 in the second level X_2 of the Demailly–Semple tower; see Proposition 6.2. We then distinguish two subcases:

- If the canonical evaluation of the restricted section along $f_{[2]}$ is nonzero, we derive a Second Main Theorem-type estimate from the intrinsic tautological evaluation and the partition-of-unity argument of [37, Theorem 3.1]; see Theorem 6.5.
- If the canonical evaluation is identically zero, Lemma 6.6 shows that the lifting $f_{[1]}(\mathbb{C})$ is algebraically degenerate in the first level X_1 of the logarithmic Demailly-Semple tower for $(\mathbb{P}^2, \mathscr{C})$, and hence McQuillan's Theorem 1.1 applies.

Case 3: The Remaining Possibilities. For the remaining pairs (m,t) not covered by the previous two cases, we establish the following Key Vanishing Lemma I to rule out the existence of any nonzero ω_1 in these settings. Note that vanishing for (m,t) implies vanishing for all (m,t') with $t' > t$. Thus we complete the proof of a Second Main Theorem for three generic conics with a finite constant c. Moreover, careful numerical computation shows that we can take $c = 19$; see Subsection 6.3.

1.7. **New Vanishing Results and Proof Strategy.**

Key Vanishing Lemma I. *For a generic configuration of three conics* $\mathscr{C} = \cup_{i=1}^{3} \mathscr{C}_i$ *in* $\mathbb{P}^2$*, one has:*

$$H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big) = 0$$

for all (m,t) *in the set* $\{(3,3),(4,4),(5,5),(6,6),(7,7)\}$.

While such vanishing results have long been recognized as fundamental in complex hyperbolicity theory, a proof has remained out of reach for lack of a viable method. The decisive innovation is a new exponent bound that converts the a priori infinite extension problem into finitely many exact linear-algebra calculations.

The project began with the question of whether three conics admit an effective Second Main Theorem and with an attempt to use slanted vector fields of pole order 3. After substantial work, that route did not yield a complete argument; moreover, lowering the pole order would not by itself remove the Key Vanishing obstruction. The present proof therefore uses the classical pole-order-7 package and resolves the remaining finite obstruction through the new exponent bound.

We later observed, with the benefit of the pole-lowering ideas developed in the subsequent papers [14, 36], that a pole-order-4 version can also be made to close. We do not pursue that variant here: the pole-order-7 argument keeps the exposition focused on the genuinely new ingredient of the present paper, namely the exponent bound and the resulting finite Key Vanishing problem. Lowering the pole order changes the finite list, but does not eliminate the need to prove its members.

Each case in Key Vanishing Lemma I follows from the corresponding stronger case in Key Vanishing Lemma II by twist monotonicity. The substantive finite verification is therefore the proof of Key Vanishing Lemma II, presented in Subsection 7.2. It has three steps:

- **All-layer finite-dimensional reduction:** The exponent bound is established uniformly in the Wronskian layer q. It restricts each coefficient to a finite-dimensional space of homogeneous polynomials, including the coefficients in all layers that can occur when m is large.
- **Exact C++ verification:** For explicit Fermat-type conics, a C++ program constructs the complete coupled transition matrix over $\mathbb{F}_5$ and proves full column rank by exact sparse elimination.
- **Geometric extension:** Upper semicontinuity globalizes the verified vanishing from this explicit configuration to a nonempty Zariski-open set of triples of conics.

Remark 1.10. An alternative approach to establishing such vanishing results is through cohomological diagram chasing based on the filtration structure (2.5). This method enables systematic construction of long exact sequences in cohomology, thereby facilitating stepwise computation of $H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}(-t)\big)$. When $m = t$ is not divisible by 3, the vanishing reduces to negatively twisted logarithmic 1-jet differentials. This observation is due to Pengchao Wang; see Appendix C. Proposition 1.5 supplies all the required symmetric-power vanishings. Consequently, Proposition C.2 gives, in particular, the cases $(m,t) = (4,4),(5,5),(7,7)$ in Key Vanishing Lemma I.

However, in more general cases—such as when $m > t$ or when 3 divides $m = t$—this diagram-chasing approach becomes substantially more complicated and its feasibility remains uncertain. Determining the essential information required for the diagram chasing presents a significant challenge. In particular, computing H^i for $i = 0,1,2$ for each term in the long exact sequence requires either analyzing the connecting homomorphisms from H^i to H^{i+1}, or computing other H^i groups that typically do not vanish, thereby introducing considerable additional complexity.

Remark 1.11. The significance of the present method extends well beyond the numerical constant in the Second Main Theorem. To the best of our knowledge, this is the first effective Second Main Theorem in higher-dimensional Nevanlinna theory whose decisive input is a pair of algebraically independent, negatively twisted invariant logarithmic 2-jet differentials, rather than a Cartan-type Wronskian construction. Here the distinction concerns the proof mechanism: the local algebra of invariant 2-jet differentials still contains its intrinsic Wronskian generator, but the Second Main Theorem is not produced by the classical Cartan–Wronskian method. This introduces a new algebraic-geometric mechanism for effectiveness. Equally importantly, the exponent bound introduced here reduces the extension problem to finite exact algebra and became a foundational input in the subsequent papers [35, 14]. In those works, the compact threshold case $d = 15$ and the logarithmic threshold case $d = 11$ are reduced to explicit finite rank-verification problems. The corresponding computer calculations are now in progress; no further conceptual ingredient is required, so reaching the two threshold degrees predicted by the 2-jet method is now a matter of computation time.

1.8. **Proof Blueprint of Theorem 1.6 with $c = 5$.** We first show that in the existence result for ω_1 in (1.8), the ratio t/m can be made close to $\frac{4-\sqrt{10}}{3}$; see Proposition 4.2. If $f^*\omega_1 \not\equiv 0$, then Theorem 1.2 yields the estimate (1.7) with constant c around $\frac{3}{4-\sqrt{10}} = 3.58\ldots$. This value represents the theoretical limit of our current approach. Our goal of achieving $c = 5$ is strategically set near this limit.

In **Case 1** ($t > 7$), the original differential ω_1 gives the coefficient m/t, while the global differential ω_2 produced by a pole-order-7 slanted vector field gives the coefficient $m/(t-7)$. Requiring both to be at most 5 is equivalent to $t \geqslant m/5+7$.

In **Case 2** ($t < m$), we refine Demailly–El Goul's approach [18] to estimate the ratio of the vanishing order of ω_2 to its weighted degree. Through the analysis in Section 6, we show that the restricted-section argument gives any constant $c > 3/\tau_1(\frac{t}{m})$, where

$$\tau_1(\tfrac{t}{m}) = \frac{3}{2}\left(4-\frac{t}{m}\right) - \frac{\sqrt{3}}{2}\sqrt{32-8\frac{t}{m}+3\left(\frac{t}{m}\right)^2}.$$

A direct computation shows that $3/\tau_1(\frac{t}{m}) < 5$ is equivalent to $t/m < 43/85$.

Although most positive integer pairs (m,t) satisfy either $t/m < 43/85$ or $t \geqslant m/5+7$, exceptional cases remain where neither condition holds. To address these, we need the following strengthened vanishing results:

Key Vanishing Lemma II. *For a generic configuration of three conics $\mathscr{C} = \cup_{i=1}^3 \mathscr{C}_i$ in $\mathbb{P}^2$, one has:*

$$H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big) = 0 \tag{1.9}$$

for all

$$\begin{aligned}(m,t) \in \big\{&(3,2),(4,3),(5,3),(6,4),(7,4),(8,5),(9,5),\\ &(10,6),(11,6),(12,7),(13,7),(14,8),(15,8),(16,9),\\ &(17,9),(18,10),(19,10),(21,11)\big\}.\end{aligned}$$

The proof in Subsection 7.2 first establishes the uniform all-layer finite shape. It then specializes to Fermat-type conics and proves, by exact full-coupled sparse elimination over $\mathbb{F}_5$, that the resulting integral matrix has trivial kernel.

1.9. **Structure of the article.** Section 2 provides necessary background. Section 3 presents detailed Riemann–Roch computations that form the core of the proof of Theorem 1.3. Section 4 advances to more sophisticated Riemann–Roch computations, specifically designed to establish the existence of negatively twisted invariant logarithmic 2-jet differentials for three generic conics. Section 5 details the application of Siu's slanted vector fields in **Case 1**. Section 6 then introduces a refinement of the Demailly–El Goul argument, adapted to handle **Case 2**. The main innovation of the article is contained in Section 7, where

we overcome fundamental obstructions in our low-degree setting by establishing Key Vanishing Lemmas. Appendix A records the explicit quotient polynomials in the four factorization identities, Appendix B proves the one-jet vanishing theorem, and Appendix C presents Pengchao Wang's cohomological observation.

2. Preliminaries

2.1. **Logarithmic Jet Bundles and Invariant Differential Forms.** Let X be an n-dimensional complex manifold. The k-th jet bundle $J_k(X)$ is defined by considering holomorphic germs $f\colon(\mathbb{C},0)\to(X,x)$, where two germs are considered equivalent if their Taylor expansions coincide up to order k in some local coordinate system. The equivalence class of a germ f, denoted by $j_k(f)$, is said to be regular if $f'(0)\neq 0$.

The total space $J_k(X)=\cup_{x\in X}J_k(X)_x$ carries a natural topology and, together with the natural projection $\pi_k\colon J_k(X)\to X$, forms a holomorphic fiber bundle. Note that $J_1(X)$ coincides with the tangent bundle T_X.

Now let $D\subset X$ be a normal crossing divisor. The logarithmic cotangent bundle $T^*_X(\log D)$ is a locally free sheaf generated by meromorphic 1-forms with at most logarithmic poles along D. Precisely, for $x\in X\setminus D$, the stalk $T^*_{X,x}(\log D)$ coincides with the stalk $T^*_{X,x}$ of the usual cotangent sheaf T^*_X at x. For $x\in D$, take a neighborhood U of x and irreducible holomorphic functions $\eta_1,\ldots,\eta_\ell$ on U such that $D\cap U=\{\eta_1\cdots\eta_\ell=0\}$; then the stalk is given by $T^*_{X,x}(\log D)=\sum_{j=1}^{\ell}\mathscr{O}_{X,x}\frac{\mathrm{d}\eta_j}{\eta_j}+T^*_{X,x}$. Its dual, the logarithmic tangent bundle $T_X(-\log D)$, consists of holomorphic vector fields tangent to each component of D (cf. [50, 25]).

Consider an open subset $U\subset X$, a section $\omega\in H^0(U,T^*_X(\log D))$. For a holomorphic curve germ $f:(\mathbb{C},0)\to U\setminus D$, write

$$f^*\omega=A_f(t)\,dt.$$

Since ω is holomorphic on $U\setminus D$, the quantities $A_f^{(j)}(0)$, for $0\le j\le k-1$, depend only on the ordinary k-jet $j_k(f)$. Thus they define a holomorphic map

$$\widetilde{\omega}:J_k(U\setminus D)\longrightarrow\mathbb{C}^k,\qquad j_k(f)\longmapsto\big(A_f(0),A_f'(0),\ldots,A_f^{(k-1)}(0)\big). \tag{2.1}$$

In particular, for every logarithmic one-form ω on U, the map in (2.1) has a unique holomorphic extension

$$\widetilde{\omega}:J_k(X,-\log D)|_U\longrightarrow\mathbb{C}^k. \tag{2.2}$$

Here the extension is taken on the logarithmic jet bundle, rather than on the ordinary jet bundle.

A holomorphic section $s:U\to J_kX|_U$ is called a *logarithmic k-jet field* if, for every open subset $V\subset U$ and every $\omega\in H^0(V,T^*_X(\log D))$, the map

$$\widetilde{\omega}\circ s|_{V\setminus D}:V\setminus D\longrightarrow\mathbb{C}^k$$

extends holomorphically to V. Such an extension, when it exists, is unique. The resulting sheaf is the sheaf of holomorphic sections of the logarithmic k-jet bundle along D, denoted by $J_k(X,-\log D)$ (cf. [50, 25, 58]).

Furthermore, given a local basis $\omega_1,\ldots,\omega_n$ of $T^*_X(\log D)$ on U, we obtain a trivialization:

$$J_k(X,-\log D)|_U\to U\times(\mathbb{C}^k)^n,\quad \sigma\mapsto\big(\pi_k\circ\sigma;\,\tilde{\omega}_1\circ\sigma,\ldots,\tilde{\omega}_n\circ\sigma\big).$$

A logarithmic jet differential of order k and weighted degree m at a point $x\in X$ is a polynomial $Q(f^{(1)},\ldots,f^{(k)})$ on the fiber of $J_k(X,-\log D)$ over x, satisfying the weighted homogeneity condition:

$$Q\big(\lambda f',\lambda^2 f'',\ldots,\lambda^k f^{(k)}\big)=\lambda^m Q\big(f',f'',\ldots,f^{(k)}\big),$$

for all $\lambda\in\mathbb{C}^*$ and all $(f',f'',\ldots,f^{(k)})\in J_k(X,-\log D)_x$.

The vector space $E^{GG}_{k,m}T^*_X(\log D)_x$ consists of all such jet differential polynomials defined at the point $x\in X$. Then the total space

$$E^{GG}_{k,m}T^*_X(\log D):=\cup_{x\in X}E^{GG}_{k,m}T^*_X(\log D)_x,$$

acquires the structure of a holomorphic vector bundle over X by Faà di Bruno's formula (cf. e.g. [12, 44]), referred to as the logarithmic Green–Griffiths bundle of jet differentials of order k and weighted degree m.

Let $\mathbb{G}_k$ denote the group of germs of k-jets of biholomorphisms of $(\mathbb{C},0)$:

$$\mathbb{G}_k:=\Big\{\varphi\colon t\mapsto\sum_{j=1}^{k}c_jt^j\;\Big|\;c_1\in\mathbb{C}^*,\ c_j\in\mathbb{C}\text{ for }j\geqslant 2\Big\}, \tag{2.3}$$

where the group operation is defined by composition modulo t^{k+1}. This group acts naturally on $J_k(X)$ via the right action:$\varphi\cdot j_k(f):=j_k(f\circ\varphi)$.

A logarithmic jet differential operator $P \in \Gamma\big(U, E^{GG}_{k,m}T^*_X(\log D)\big)$ over an open set $U \subset X$ is said to be invariant under the reparameterization group $\mathbb{G}_k$ if for every $\varphi \in \mathbb{G}_k$ the following identity holds:

$$P\big((f\circ\varphi)', (f\circ\varphi)'', \ldots, (f\circ\varphi)^{(k)}\big) = (\varphi')^m \cdot P\big(f', f'', \ldots, f^{(k)}\big). \tag{2.4}$$

The subbundle $E_{k,m}T^*_X(\log D) \subset E^{GG}_{k,m}T^*_X(\log D)$ is then defined as the holomorphic vector bundle whose fibers consist of all such invariant logarithmic jet differential polynomials.

This notion was initially introduced by Demailly [17] in the compact case, and subsequently generalized to the logarithmic case by Dethloff-Lu [20].

Throughout this article, we only concern the situation $k = 1$ or 2. In the case that $\dim_{\mathbb{C}} X = 2$, the bundle of invariant logarithmic jet differentials of order 2 admits a filtration structure with the grading [17]:

$$\mathrm{Gr}^\bullet E_{2,m}T^*_X(\log D) = \bigoplus_{0\leqslant j\leqslant \lfloor m/3\rfloor} S^{m-3j}T^*_X(\log D)\otimes \overline{K}^j_X, \tag{2.5}$$

where $\overline{K}_X$ is the logarithmic canonical line bundle $K_X \otimes \mathscr{O}_X(D)$. This filtration corresponds to the expression of an invariant logarithmic 2-jet differential $P(f', f'')$ outside D as:

$$P(f', f'') = \sum_{0\leqslant j\leqslant \lfloor m/3\rfloor} \sum_{\substack{\alpha_1+\alpha_2=m-3j\\ \alpha_1,\alpha_2\geqslant 0}} R_{\alpha_1,\alpha_2,j}(f_1,f_2)\cdot (f_1')^{\alpha_1}(f_2')^{\alpha_2}(f_1'f_2''-f_2'f_1'')^j,$$

where $f = (f_1, f_2)$ represents the holomorphic germs in local coordinates, and where $R_\bullet$ are holomorphic functions.

2.2. **Logarithmic Demailly-Semple Tower.** The main reference of this subsection is [17, 20].

For simplicity, consider the log-direct manifold (X_0, D_0, V_0) associated with three generic conics $\mathscr{C}_1$, $\mathscr{C}_2$, $\mathscr{C}_3$ in the complex projective plane $\mathbb{P}^2$, where $X_0 = X := \mathbb{P}^2$, $D_0 := \mathscr{C} = \sum_{i=1}^3 \mathscr{C}_i$ is a simple normal crossing divisors on X whose irreducible components $\mathscr{C}_i$ are smooth conics and $V_0 := T_X(-\log D_0)$. Define $X_1 := \mathbb{P}(V_0)$ with the natural projection $\pi_{1,0} : X_1 \to X_0$. Set $D_1 := \pi^*_{1,0}D_0$. The subbundle $V_1 \subset T_{X_1}(-\log D_1)$ is defined fiberwise by:

$$V_{1,(x,[v])} := \{\xi \in T_{X_1,(x,[v])}(-\log D_1) \,|\, (\pi_{1,0})_*\,\xi \in \mathbb{C}\cdot v\}, \tag{2.6}$$

with $\mathbb{C}\cdot v \subset V_{0,x} = T_{X_0,x}(-\log D_0)$. Thus we obtain the first level (X_1, D_1, V_1) of the logarithmic Demailly-Semple tower for $(\mathbb{P}^2, \mathscr{C})$.

Proceeding likewise, we can construct the second level of logarithmic Demailly–Semple tower:

$$(X_2, D_2, V_2) \to (X_1, D_1, V_1) \to (X_0, D_0, V_0),$$

together with the projections $\pi_{2,1} : X_2 \to X_1$ and $\pi_{2,0} : X_2 \to X_0$.

Let $\mathscr{O}_{X_1}(-1)$ denote the tautological line bundle of $X_1 = \mathbb{P}(V_0)$ such that $\mathscr{O}_{X_1}(-1)_{(x,[v])} = \mathbb{C}\cdot v$. By definition, there is a short exact sequence

$$0 \longrightarrow T_{X_1/X_0} \longrightarrow V_1 \xrightarrow{(\pi_{1,0})_*} \mathscr{O}_{X_1}(-1) \longrightarrow 0\ , \tag{2.7}$$

where T_{X_1/X_0} is the relative tangent bundle of the fibration $\pi_{1,0} : X_1 \to X_0$. Moreover, there is a relative version of the Euler sequence of the tangent bundle of the fiber $\mathbb{P}(V_{0,x})$:

$$0 \longrightarrow \mathscr{O}_{X_1} \longrightarrow \pi^*_{1,0}V_0 \otimes \mathscr{O}_{X_1}(1) \longrightarrow T_{X_1/X_0} \longrightarrow 0\ . \tag{2.8}$$

Furthermore, there exists a canonical injection $\mathscr{O}_{X_2}(-1) \hookrightarrow \pi^*_{2,1}V_1$. Combining this with the above sequence yields a natural morphism of line bundles:

$$\mathscr{O}_{X_2}(-1) \lhook\joinrel\longrightarrow \pi^*_{2,1}V_1 \xrightarrow{\pi^*_{2,1}(\pi_{1,0})_*} \pi^*_{2,1}\mathscr{O}_{X_1}(-1)\ ,$$

whose zero divisor is the hyperplane subbundle:

$$\Gamma_2 := \mathbb{P}(T_{X_1/X_0}) \subset \mathbb{P}(V_1) = X_2. \tag{2.9}$$

Consequently, we can obtain the isomorphism:

$$\mathscr{O}_{X_2}(\Gamma_2) \cong \pi^*_{2,1}\mathscr{O}_{X_1}(-1)\otimes \mathscr{O}_{X_2}(1). \tag{2.10}$$

Let $f\colon \mathbb{D}_R \to X_0$ be a nonconstant holomorphic map from the disc $\mathbb{D}_R = \{z \in \mathbb{C} : |z| < R\}$ of radius $R \in \mathbb{R}_+ \cup \{\infty\}$ satisfying $f(\mathbb{D}_R) \not\subset D_0$. To define its liftings to the first level X_1 of the logarithmic Demailly–Semple tower, we first consider the natural morphism

$$\rho : T_{X_0}(-\log D_0) \longrightarrow T_{X_0}$$

is an isomorphism over $X_0 \setminus D_0$. On $\mathbb{D}_R \setminus f^{-1}(D_0)$, define

$$v_f = \rho^{-1}(f').$$

With respect to a local parameter t on $\mathbb{D}_R$ and adapted coordinates satisfying $D = \{z_1 \cdots z_q = 0\}$, the local expression of v_f is

$$v_f = \left(\frac{(z_1 \circ f)'}{z_1 \circ f}, \ldots, \frac{(z_q \circ f)'}{z_q \circ f}, (z_{q+1} \circ f)', \ldots, (z_2 \circ f)' \right).$$

Then v_f extends meromorphically across the points of $f^{-1}(D)$. Locally at any point of $\mathbb{D}_R$, write

$$v_f = t^{\nu} b(t), \qquad b(t) \text{ holomorphic}, \qquad b(0) \neq 0,$$

by removing the common lowest-order factor. The regularized first lift is defined by

$$f_{[1]}(t) = (f(t); [b(t)]) \in X_1.$$

where $[b(t)]$ agrees with $[v_f]$ wherever the latter is directly defined and is independent of the choices involved. With this convention, the first lifting $f_{[1]}$ of f is a well-defined holomorphic map from $\mathbb{D}_R$ to X_1. Repeating the process for $f_{[1]}$, the second lifting $f_{[2]}$ of f is also well-defined and holomorphic from $\mathbb{D}_R$ to X_2.

Moreover, from the definition of Γ_2 and the identity $\pi_{1,0} \circ f_{[1]} = f$, at every point t such that $f(t) \notin D_0$ and $f'(t) \neq 0$, one has $f_{[2]}(t) \notin \Gamma_2$.

We next explain the corresponding construction of sections. Let $m \geq 0$ and

$$P \in H^0\big(U, E_{2,m} T^*_X(\log D)\big).$$

Write

$$\alpha_2 : J_2(X, -\log D)^{\mathrm{reg}} \longrightarrow X_2^{\mathrm{reg}}$$

for the quotient map and

$$\widetilde{\alpha}_2 : J_2(X, -\log D)^{\mathrm{reg}} \longrightarrow \mathscr{O}_{X_2}(-1)|_{X_2^{\mathrm{reg}}}$$

for its tautological lift, whose values are nonzero. Over $X \setminus D$, the latter sends $j_2(f)$ to $f'_{[1]}(0)$; over D, it is defined by the holomorphic logarithmic jet construction.

On $X_2^{\mathrm{reg}} \cap \pi_{2,0}^{-1}(U)$, define σ_P by

$$\big\langle \sigma_P(\alpha_2(\xi)), \widetilde{\alpha}_2(\xi)^{\otimes m} \big\rangle = P(\xi).$$

The transformation laws

$$P(\xi \circ \varphi) = \varphi'(0)^m P(\xi), \qquad \widetilde{\alpha}_2(\xi \circ \varphi) = \varphi'(0) \widetilde{\alpha}_2(\xi)$$

make this definition independent of the representative. The local holomorphic sections of the logarithmic quotient map show that σ_P is holomorphic, including at regular tower points over D.

Over $\pi_{2,0}^{-1}(U \setminus D)$, the ordinary Demailly–Semple correspondence gives a holomorphic section of $\mathscr{O}_{X_2}(m)$, including along Γ_2 in this open set. It agrees with the preceding construction on their common domain. Consequently, σ_P is defined and holomorphic away from $D_2 \cap \Gamma_2$. Since X is a surface, $\Gamma_2 \to X_1$ is a section of the projective bundle $X_2 \to X_1$, and $D_2 \cap \Gamma_2$ corresponds to the divisor $D_1 \subset X_1$. Hence this intersection has codimension two in X_2. As X_2 is smooth, σ_P extends uniquely across this set as a holomorphic section of $\mathscr{O}_{X_2}(m)$.

More generally, the following fundamental isomorphism holds:

Direct image formula (cf. [17, Theorem 6.8], [20, Proposition 3.9])**.** *The direct image sheaf*

$$(\pi_{2,0})_* \mathscr{O}_{X_2}(m) \cong \mathscr{O}\left(E_{2,m} T^*_{\mathbb{P}^2}(\log \mathscr{C})\right) \tag{2.11}$$

is isomorphic to the sheaf of invariant logarithmic 2*-jet differentials of weighted degree m.*

2.3. **A Bogomolov-type Vanishing Theorem.** In the subsequent Sections 3 and 4, the existence of negatively twisted logarithmic 1-jet and 2-jet differentials will be established via Riemann–Roch calculations. This reduces to verifying the vanishing of the corresponding second cohomology groups:

$$H^2\big(\mathbb{P}^2, S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big) \quad \text{and} \quad H^2\big(\mathbb{P}^2, E_{2,m} T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big),$$

respectively. A key tool in proving these vanishing results is the following Bogomolov-type vanishing theorem.

Theorem 2.1 (cf. [8, Theorem 4.4])**.** *Let (X,D) be a log pair where X is a smooth projective surface and D is a reduced simple normal crossing divisor. Assume that (X,D) is of general type (i.e., K_X+D is big) and K_X+D is nef. Then*

$$H^0\big(X, S^p T_X(-\log D) \otimes \mathscr{O}_X(q(K_X+D))\big) = 0,$$

for all integers p,q satisfying $p > 2q$.

In the compact case, Bogomolov [4] established a seminal vanishing theorem under the assumption that tangent bundle T_X is semi-stable. Later, Enoki [26] showed that T_X is K_X-semi-stable for minimal Kähler space X with big K_X, and Tsuji [68] proved that T_X is K_X-semi-stable for smooth minimal algebraic variety (i.e., K_X is nef). Combining these results, the Bogomolov-type vanishing theorem holds for projective variety X with big and nef K_X; see [17, Theorem 14.1].

3. Proof of Theorem 1.3

Let $\mathscr{C} = \sum_{i=1}^3 \mathscr{C}_i$ be a simple normal crossing divisor of total degree $d = d_1+d_2+d_3$, where $d_i = \deg \mathscr{C}_i$ satisfy $d_1 \geqslant d_2 \geqslant d_3 \geqslant 2$ and $d_1 \geqslant 3$. To derive an effective Second Main Theorem, we seek nontrivial global sections of the twisted sheaf

$$S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(1)^{-(d-3)\delta m} \cong S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}_{\mathbb{P}^2}^{-\delta m},$$

where $\delta > 0$ is a small rational number to be chosen, and m is taken sufficiently divisible.

Definition 3.1 (cf. [18, p. 517], [25, Definition 1.3.1])**.** The 1-*jet log threshold* of $(\mathbb{P}^2, \mathscr{C})$ is defined by

$$\overline{\theta}_1(\mathbb{P}^2, \mathscr{C}) := \sup\left\{ \frac{t}{m} \,\middle|\, m,t > 0,\ H^0\big(\mathbb{P}^2, S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big) \neq 0 \right\}.$$

If no such pair (m,t) exists, the threshold is defined to be zero.

For a generic configuration of three conics, Proposition 1.5 gives $\overline{\theta}_1(\mathbb{P}^2, \mathscr{C}) = 0$.

Proposition 3.2 (Proof of (1.4))**.** *Let $\mathscr{C} = \cup_{i=1}^3 \mathscr{C}_i$ be a union of three smooth curves $\mathscr{C}_i$ of degrees d_i with simple normal crossings. Assume that $d_1 \geqslant d_2 \geqslant d_3 \geqslant 2$ and $d_1 \geqslant 3$. Then one has the estimate*

$$0 < \frac{1}{\overline{\theta}_1(\mathbb{P}^2, \mathscr{C})} \leqslant \frac{1}{(d-3)\delta_1},$$

where

$$\delta_1 = \frac{1}{2} - \frac{\sqrt{9(d-1)^2 - 12(d_1d_2+d_2d_3+d_3d_1)}}{6(d-3)} > 0. \tag{3.1}$$

Proof. We follow the standard Riemann-Roch argument (cf. [18, 25]).

By Serre duality, the dual of

$$H^2\big(\mathbb{P}^2, S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}_{\mathbb{P}^2}^{-\delta m}\big) \tag{3.2}$$

is

$$H^0\big(\mathbb{P}^2, K_{\mathbb{P}^2} \otimes S^m T_{\mathbb{P}^2}(-\log \mathscr{C}) \otimes \overline{K}_{\mathbb{P}^2}^{\delta m}\big).$$

Take m sufficiently divisible that δm is an integer. Multiplication by a nonzero section of $\mathscr{O}_{\mathbb{P}^2}(3)$ injects the last sheaf into

$$S^m T_{\mathbb{P}^2}(-\log \mathscr{C}) \otimes \overline{K}_{\mathbb{P}^2}^{\delta m}.$$

For $0 < \delta < 1/2$ one has $m > 2\delta m$. The Bogomolov-type Vanishing Theorem 2.1 therefore annihilates the target, and hence also the Serre-dual group. We obtain

$$H^2\big(\mathbb{P}^2, S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}_{\mathbb{P}^2}^{-\delta m}\big) = 0$$

which yields

$$\dim H^0\Big(\mathbb{P}^2, S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}^{-\delta m}_{\mathbb{P}^2}\Big) \geqslant \chi\Big(S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}^{-\delta m}_{\mathbb{P}^2}\Big),$$

for large m. A standard Riemann-Roch computation gives

$$\begin{aligned}\chi\Big(S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}^{-\delta m}_{\mathbb{P}^2}\Big) &= \frac{1}{6}\big((m-\delta m)^3 - (-\delta m)^3\big)\bar{c}_1^2 - \frac{1}{6}m^3\bar{c}_2 + O(m^2)\\ &= \frac{1}{2}m^3\left((d-3)^2\delta^2 - (d-3)^2\delta + \frac{1}{3}(d_1d_2+d_2d_3+d_3d_1) - d + 2\right) + O(m^2).\end{aligned}$$

The leading coefficient $\varphi(\delta) = (d-3)^2\delta^2 - (d-3)^2\delta + \frac{1}{3}(d_1d_2+d_2d_3+d_3d_1) - d + 2$ has two positive roots given by

$$\delta_1 = \frac{1}{2} - \frac{\sqrt{9(d-1)^2 - 12(d_1d_2+d_2d_3+d_3d_1)}}{6(d-3)}, \qquad \delta_2 = \frac{1}{2} + \frac{\sqrt{9(d-1)^2 - 12(d_1d_2+d_2d_3+d_3d_1)}}{6(d-3)} > 0.$$

We can check the positivity of δ_1 directly. Hence $\varphi(\delta) > 0$ for all $\delta \in (0, \delta_1)$. Thus we conclude the proof. □

4. Twisted Invariant Logarithmic 2-Jet Differentials and Threshold

To obtain a constant c as small as possible in the Second Main Theorem for three conics $\mathscr{C} \subset \mathbb{P}^2$, by Theorem (1.2), we seek negatively twisted invariant logarithmic 2-jet differentials with relatively high vanishing order t compared to the degree m. This motivates the study of the following threshold:

Definition 4.1 (cf. [18, p. 517], [25, Definition 1.3.1])**.** The 2-*jet log threshold* of $(\mathbb{P}^2, \mathscr{C})$ is defined by

$$\overline{\theta}_2(\mathbb{P}^2, \mathscr{C}) := \sup\left\{\frac{t}{m}\,\middle|\, m,t>0,\ H^0\big(X_2, \mathscr{O}_{X_2}(m) \otimes \pi_{2,0}^*\mathscr{O}_X(-t)\big) \neq 0\right\}.$$

If no such pair (m,t) exists, the threshold is defined to be zero.

If $\overline{\theta}_2 > 0$, then for any sufficiently small $\varepsilon > 0$, there exists a negatively twisted logarithmic 2-jet differential

$$\omega \in H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big)$$

with ratio $\overline{\theta}_2 - \varepsilon < \frac{t}{m} \leqslant \overline{\theta}_2$. Furthermore, if $f^*\omega \not\equiv 0$ also holds, then Theorem 1.2 yields the Second Main Theorem with constant

$$\frac{1}{\overline{\theta}_2} \leqslant c = \frac{m}{t} < \frac{1}{\overline{\theta}_2 - \varepsilon}.$$

The significance of the 2-jet log threshold stems from its role as the fundamental barrier that dictates the possibility of constructing jet differentials with a prescribed ratio of vanishing orders to weighted degrees.

The following result establishes an explicit upper bound for $\frac{1}{\overline{\theta}_2}$ in the case of three smooth conics with simple normal crossings:

Proposition 4.2. *Let $\mathscr{C} = \cup_{i=1}^3 \mathscr{C}_i$ be a union of three smooth conics $\mathscr{C}_i$ with simple normal crossings. Then one has the estimate*

$$0 < \frac{1}{\overline{\theta}_2(\mathbb{P}^2, \mathscr{C})} \leqslant \frac{3}{4-\sqrt{10}} = 3.58\ldots.$$

The proof employs a Riemann–Roch calculation, which is standard (cf. [18]).

Proof. We seek nonzero invariant logarithmic 2-jet differential forms with high vanishing order, which correspond to global sections of the twisted vector bundle

$$E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(1)^{-3\delta m} \cong E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}^{-\delta m}_{\mathbb{P}^2},$$

where the logarithmic canonical line bundle

$$\overline{K}_{\mathbb{P}^2} = K_{\mathbb{P}^2} \otimes \mathscr{O}(\mathscr{C}) \cong \mathscr{O}_{\mathbb{P}^2}(-3) \otimes \mathscr{O}_{\mathbb{P}^2}(6) \cong \mathscr{O}_{\mathbb{P}^2}(3),$$

with $\delta > 0$ a rational number to be chosen and m sufficiently divisible.

The standard approach begins with Serre duality:

$$H^2\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}^{-\delta m}_{\mathbb{P}^2}\big)^\vee = H^0\big(\mathbb{P}^2, K_{\mathbb{P}^2} \otimes \big(E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C})\big)^\vee \otimes \overline{K}^{\delta m}_{\mathbb{P}^2}\big). \tag{4.1}$$

The bundle on the right admits the filtration induced by the dual of (2.5), with graded pieces

$$S^{m-3j}T_{\mathbb{P}^2}(-\log\mathscr{C})\otimes\overline{K}_{\mathbb{P}^2}^{-j}\otimes K_{\mathbb{P}^2}\otimes\overline{K}_{\mathbb{P}^2}^{\delta m}=S^{m-3j}T_{\mathbb{P}^2}(-\log\mathscr{C})\otimes\overline{K}_{\mathbb{P}^2}^{\delta m-j-1}$$

for $0\leqslant j\leqslant\frac{m}{3}$.

For any rational number $\delta\in(0,\frac{1}{3})$ and sufficiently divisible $m\gg 1$, the inequality

$$m-3j>2(\delta m-j-1)$$

automatically holds for all $0\leqslant j\leqslant\frac{m}{3}$. By applying the Bogomolov-type Vanishing Theorem 2.1, we deduce:

$$H^0\Big(\mathbb{P}^2,S^{m-3j}T_{\mathbb{P}^2}(-\log\mathscr{C})\otimes\overline{K}_{\mathbb{P}^2}^{\delta m-j-1}\Big)=0\quad\text{ for all }\quad 0\leqslant j\leqslant\tfrac{m}{3}.$$

This establishes the vanishing of the right-hand side in (4.1), which consequently implies the vanishing of the left-hand side as well. We thus obtain the lower bound

$$\dim H^0\Big(\mathbb{P}^2,E_{2,m}T^*_{\mathbb{P}^2}(\log\mathscr{C})\otimes\overline{K}_{\mathbb{P}^2}^{-\delta m}\Big)\geqslant\chi\Big(E_{2,m}T^*_{\mathbb{P}^2}(\log\mathscr{C})\otimes\overline{K}_{\mathbb{P}^2}^{-\delta m}\Big),$$

since the Euler characteristic decomposes as

$$\chi(\cdots)=\dim H^0(\cdots)-\dim H^1(\cdots)+\dim H^2(\cdots)=\dim H^0(\cdots)-\dim H^1(\cdots).$$

To estimate the asymptotic growth of the Euler characteristic, we employ the Schur bundle filtration (see (2.5)):

$$\begin{aligned}\mathrm{Gr}^\bullet E_{2,m}T^*_{\mathbb{P}^2}(\log\mathscr{C})&=\bigoplus_{a+3b=m}\Gamma^{(a+b,b)}T^*_{\mathbb{P}^2}(\log\mathscr{C})\qquad\text{(cf. [17, p. 67], [25, p. 474])}\\&=\bigoplus_{0\leqslant b\leqslant\frac{m}{3}}\Gamma^{(m-2b,b)}T^*_{\mathbb{P}^2}(\log\mathscr{C}).\end{aligned}\tag{4.2}$$

Together with the Pieri formula (cf. [28, p. 455], [17, p. 66]):

$$\Gamma^{(\ell_1,\ell_2)}T^*_{\mathbb{P}^2}(\log\mathscr{C})\otimes\overline{K}_{\mathbb{P}^2}^{-\delta m}=\Gamma^{(\ell_1-\delta m,\ell_2-\delta m)}T^*_{\mathbb{P}^2}(\log\mathscr{C}),$$

we derive the following computation (cf. [44, p. 77]):

$$\begin{aligned}&\chi\big(E_{2,m}T^*_{\mathbb{P}^2}(\log\mathscr{C})\otimes\overline{K}_{\mathbb{P}^2}^{-\delta m}\big)\\&=\sum_{b=0}^{\frac{m}{3}}\chi\big(\Gamma^{(m-2b-\delta m,b-\delta m)}T^*_{\mathbb{P}^2}(\log\mathscr{C})\big)\\&=\left(\frac{1}{6}\int_0^{\frac{m}{3}}\big[(m-2b-\delta m)^3-(b-\delta m)^3\big]\mathrm{d}b\right)\bar{c}_1^2-\left(\frac{1}{6}\int_0^{\frac{m}{3}}(m-3b)^3\mathrm{d}b\right)\bar{c}_2+O(m^3)\\&=\frac{m^2(m+3)}{648}(54\delta^2m-48\delta m+13m-18\delta+15)\bar{c}_1^2-\frac{m^4}{72}\bar{c}_2+O(m^3)\\&=\frac{m^4}{648}\big((54\delta^2-48\delta+13)\bar{c}_1^2-9\bar{c}_2\big)+O(m^3).\end{aligned}$$

Here the logarithmic Chern numbers are determined by the formula $\bar{c}_1^2=(d-3)^2=9$ and $\bar{c}_2=d^2-3d+3-\sum_{i<j}d_id_j=9$, where $d_1=d_2=d_3=2$ and $d=d_1+d_2+d_3=6$.

The quadratic leading coefficient (after removing the common factor $\frac{9}{648}$)

$$\varphi(\delta):=(54\delta^2-48\delta+13)-9=54\delta^2-48\delta+4$$

has two positive roots given by $\delta_1=\frac{4-\sqrt{10}}{9}$ and $\delta_2=\frac{4+\sqrt{10}}{9}$.

Hence, $\varphi(\delta)>0$ for all $\delta\in(0,\delta_1)$. It follows that $\overline{\theta}_2\geqslant 3\delta_1=0.279\ldots$, which completes the proof of Proposition 4.2. □

Siu's strategy of slanted vector fields requires families of negatively twisted jet differentials; see [65, pp. 558–559]. In Proposition 4.3, we will use algebraic cohomology and base change on the relative logarithmic tower to extend the section defining the residual divisor.

Now we apply Demailly–El Goul's argument (cf. [18, p. 532], [25, Lemma 1.3.2]) to show the existence of "irreducible" negatively twisted invariant logarithmic 2-jet differentials.

Proposition 4.3. *Let $S=(\mathbb{P}^5)^3$ be the parameter space of ordered triples of plane conics, and let $S^{\mathrm{snc}}\subset S$ be the nonempty Zariski-open subset parametrizing triples of smooth conics whose union $\mathscr{C}_a=\mathscr{C}_{1,a}+\mathscr{C}_{2,a}+\mathscr{C}_{3,a}$ is a simple normal crossing divisor. There exist positive integers m,t, integers b_1,b_2 with*

$$b_1\geqslant 2b_2>0,\qquad b_1+b_2=m,\qquad \frac{t}{m}>\frac{1}{5},$$

and a nonempty Zariski-open subset $U^{m,t}\subset S^{\mathrm{snc}}$ on which there is an algebraic, hence holomorphic, family of nonzero invariant logarithmic 2-jet differentials

$$\omega_{1,a}\in H^0\big(\mathbb{P}^2,E_{2,m}T^*_{\mathbb{P}^2}(\log\mathscr{C}_a)\otimes\mathscr{O}_{\mathbb{P}^2}(-t)\big),\qquad a\in U^{m,t}.$$

Viewed as a section on the second logarithmic Demailly–Semple stage $X_{2,a}$ via (2.11), each $\omega_{1,a}$ has zero divisor

$$\operatorname{div}(\omega_{1,a})=Z_a+b_1\Gamma_{2,a},$$

where Z_a is irreducible and reduced and $Z_a\neq\Gamma_{2,a}$. In particular,

$$Z_a\sim b_1u_{1,a}+b_2u_{2,a}-t\pi^*_{2,0}h,$$

where $u_{1,a}$ and $u_{2,a}$ denote the classes of $\pi^*_{2,1}\mathscr{O}_{X_{1,a}}(1)$ and $\mathscr{O}_{X_{2,a}}(1)$, respectively, and h denotes the class of $\mathscr{O}_{\mathbb{P}^2}(1)$.

Proof. We first construct the logarithmic Demailly–Semple tower relatively over S^{snc}. Let $\mathscr{C}\subset\mathbb{P}^2\times S^{\mathrm{snc}}$ be the universal relative simple normal crossing divisor (see (5.1)). The relative logarithmic tangent bundle

$$T_{\mathbb{P}^2\times S^{\mathrm{snc}}/S^{\mathrm{snc}}}(-\log\mathscr{C})$$

is locally free of rank two. Its relative tower consists of projective bundles

$$\mathscr{X}_2\xrightarrow{\Pi_{2,1}}\mathscr{X}_1\xrightarrow{\Pi_{1,0}}\mathbb{P}^2\times S^{\mathrm{snc}},$$

with a smooth projective structure morphism $q\colon\mathscr{X}_2\to S^{\mathrm{snc}}$ and a relative vertical divisor Γ_2. Their fibres are $X_{2,a}$, $X_{1,a}$ and $\Gamma_{2,a}$, respectively. We write $\Pi_{2,0}=\Pi_{1,0}\circ\Pi_{2,1}$ and suppress the projection to $\mathbb{P}^2$ when pulling back line bundles from the base surface. For $(c_1,c_2,s)\in\mathbb{Z}^3$, set

$$\mathscr{L}_{c_1,c_2,s}:=\Pi^*_{2,1}\mathscr{O}_{\mathscr{X}_1}(c_1)\otimes\mathscr{O}_{\mathscr{X}_2}(c_2)\otimes\Pi^*_{2,0}\mathscr{O}_{\mathbb{P}^2}(-s).$$

Generic cohomology and base change for this projective flat family (cf. [33, Chapter III, Section 12]) gives a dense Zariski-open subset $S_{c_1,c_2,s}\subset S^{\mathrm{snc}}$ on which $q_*\mathscr{L}_{c_1,c_2,s}$ is locally free and the natural maps

$$(q_*\mathscr{L}_{c_1,c_2,s})\otimes\mathbb{C}(a)\longrightarrow H^0(X_{2,a},\mathscr{L}_{c_1,c_2,s}|_{X_{2,a}})$$

are isomorphisms. Since there are only countably many integer triples and S^{snc} is irreducible over the uncountable field $\mathbb{C}$, we may choose a complex point

$$a_0\in\bigcap_{(c_1,c_2,s)\in\mathbb{Z}^3}S_{c_1,c_2,s}$$

which also lies in the generic locus of Proposition 1.5.

Proposition 4.2 gives

$$\overline{\theta}_2(\mathbb{P}^2,\mathscr{C}_{a_0})\geqslant\frac{4-\sqrt{10}}{3}>\frac{1}{5}.$$

By the definition of the threshold and (2.11), there exist positive integers M,T with $T/M>1/5$ and a nonzero section

$$\omega_0\in H^0\Big(X_{2,a_0},\mathscr{O}_{X_{2,a_0}}(M)\otimes\pi^*_{2,0}\mathscr{O}_{\mathbb{P}^2}(-T)\Big).$$

Write $\operatorname{div}(\omega_0)=\sum_j p_jZ_j$, where the Z_j are distinct irreducible reduced divisors and $p_j\geqslant 1$. With $u_i=u_{i,a_0}$, the projective-bundle formula for the Picard group gives

$$\operatorname{Pic}(X_{2,a_0})=\mathbb{Z}\pi^*_{2,0}h\oplus\mathbb{Z}u_1\oplus\mathbb{Z}u_2.$$

Thus we can write

$$Z_j\sim b_{1,j}u_1+b_{2,j}u_2-t_j\pi^*_{2,0}h,\qquad b_{1,j},b_{2,j},t_j\in\mathbb{Z}.$$

The divisor-class argument of [18, Lemma 3.3(d)], applied to the logarithmic tower as in [25, Lemma 1.3.2], gives the following three cases:

- $(b_{1,j},b_{2,j})=(0,0)$ and $t_j<0$; in this case Z_j is pulled back from a curve in $\mathbb{P}^2$;
- $Z_j=\Gamma_{2,a_0}$, with $(b_{1,j},b_{2,j},t_j)=(-1,1,0)$;

- $b_{1,j} \geqslant 2b_{2,j} \geqslant 0$ and $m_j := b_{1,j} + b_{2,j} > 0$.

In the third case, the relation

$$\pi_{2,1}^{*}\mathscr{O}_{X_{1,a_0}}(b_1) \otimes \mathscr{O}_{X_{2,a_0}}(b_2) \cong \mathscr{O}_{X_{2,a_0}}(b_1+b_2) \otimes \mathscr{O}_{X_{2,a_0}}(-b_1\Gamma_{2,a_0}), \quad \text{see (2.10)}$$

gives

$$Z_j \sim b_{1,j}u_1 + b_{2,j}u_2 - t_j\pi_{2,0}^{*}h \sim m_j u_2 - b_{1,j}\Gamma_{2,a_0} - t_j\pi_{2,0}^{*}h.$$

Let I denote the set of indices corresponding to the third case. Since $\sum_j p_j[Z_j] = Mu_2 - T\pi_{2,0}^{*}h$, comparison of the u_1- and u_2-coefficients, yields

$$M = \sum_{j\in I} p_j m_j.$$

Indeed, divisors of the first type have zero relative coefficients, while for $Z_j = \Gamma_{2,a_0}$ the two relative coefficients sum to $-1+1=0$. Since $M>0$, it follows that $I \neq \varnothing$.

Comparing the coefficients of $\pi_{2,0}^{*}h$, we obtain

$$T = \sum_{j} p_j t_j \leqslant \sum_{j\in I} p_j t_j,$$

since the first two cases contribute only nonpositive t_j. Consequently,

$$\max_{j\in I} \frac{t_j}{m_j} \geqslant \frac{\sum_{j\in I} p_j t_j}{\sum_{j\in I} p_j m_j} \geqslant \frac{T}{M} > \frac{1}{5}.$$

Choose a component $Z = Z_j$ attaining this maximum and put $b_1 = b_{1,j}$, $b_2 = b_{2,j}$, $m = b_1 + b_2$ and $t = t_j$. Then $m, t > 0$, $b_1 \geqslant 2b_2 \geqslant 0$ and $t/m > 1/5$. The canonical section of $\mathscr{O}_{X_{2,a_0}}(Z)$, after choosing an isomorphism with $\mathscr{L}_{b_1,b_2,t}|_{X_{2,a_0}}$, gives

$$\widetilde{\omega}_{a_0} \in H^0(X_{2,a_0}, \mathscr{L}_{b_1,b_2,t}|_{X_{2,a_0}}), \qquad \operatorname{div}(\widetilde{\omega}_{a_0}) = Z.$$

If $b_2 = 0$, then $b_1 = m$. Pushing this nonzero section down the two projective bundles and applying the projection formula would give

$$0 \neq H^0\left(\mathbb{P}^2, S^m T^{*}_{\mathbb{P}^2}(\log \mathscr{C}_{a_0}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\right),$$

contrary to Proposition 1.5. Hence $b_1 \geqslant 2b_2 > 0$.

We now extend the section defining Z. Set $\mathscr{L} = \mathscr{L}_{b_1,b_2,t}$. By the choice of a_0, the sheaf $q_*\mathscr{L}$ is locally free near a_0 and commutes with base change there. After choosing a local algebraic frame, the vector $\widetilde{\omega}_{a_0}$ extends to a section of $q_*\mathscr{L}$ on a Zariski-open neighbourhood U of a_0, hence to

$$\widetilde{\omega} \in H^0(q^{-1}(U), \mathscr{L}).$$

Since $\widetilde{\omega}|_{X_{2,a_0}} = \widetilde{\omega}_{a_0} \neq 0$, the locus of points $a \in U$ for which $\widetilde{\omega}|_{X_{2,a}}$ vanishes identically is Zariski closed in U. After replacing U by a smaller Zariski-open neighbourhood of a_0, we may assume that

$$\widetilde{\omega}|_{X_{2,a}} \neq 0 \qquad \text{for every } a \in U.$$

Since each fibre $X_{2,a}$ is smooth and geometrically integral, these restrictions are nonzerodivisors. The fibrewise criterion for relative effective Cartier divisors therefore shows that

$$\mathscr{Z} := V(\widetilde{\omega}) \subset q^{-1}(U)$$

is an effective Cartier divisor flat over U. It is also proper and of finite presentation over U. Its fibre at a_0 is the irreducible reduced divisor Z, which is geometrically integral over $\mathbb{C}$. The openness of the geometrically integral fibre locus for proper flat morphisms of finite presentation [76, Theorem 12.2.1(x)] now allows us to shrink U around a_0 so that every $Z_a := \mathscr{Z}_a$ is geometrically integral. In particular, for every complex parameter $a \in U$, the divisor Z_a is irreducible and reduced, with

$$Z_a \sim b_1 u_{1,a} + b_2 u_{2,a} - t\pi_{2,0}^{*}h.$$

It cannot equal $\Gamma_{2,a}$: their classes in the projective-bundle Picard group differ, since $\Gamma_{2,a}$ has class $-u_{1,a} + u_{2,a}$ and $t > 0$.

Finally, the relative form of (2.10) is

$$\mathscr{O}_{\mathscr{X}_2}(\Gamma_2) \simeq \Pi_{2,1}^{*}\mathscr{O}_{\mathscr{X}_1}(-1) \otimes \mathscr{O}_{\mathscr{X}_2}(1).$$

Let s_{Γ_2} be its canonical section and set $\omega_1 = s_{\Gamma_2}^{b_1}\widetilde{\omega}$ on $q^{-1}(U)$. Since $m = b_1 + b_2$, this is an algebraic section of $\mathscr{O}_{\mathscr{X}_2}(m) \otimes \pi_{2,0}^* \mathscr{O}_{\mathbb{P}^2}(-t)$ and satisfies

$$\operatorname{div}(\omega_{1,a}) = Z_a + b_1 \Gamma_{2,a} \qquad (a \in U).$$

The relative direct-image correspondence underlying (2.11) identifies its restrictions with an algebraic, hence holomorphic, family

$$0 \neq \omega_{1,a} \in H^0\big(\mathbb{P}^2, E_{2,m} T^*_{\mathbb{P}^2}(\log \mathscr{C}_a) \otimes \mathscr{O}_{\mathbb{P}^2}(-t)\big), \qquad \frac{t}{m} > \frac{1}{5}. \tag{4.3}$$

Taking $U^{m,t} = U$ proves the proposition. □

5. Generating Independent Twisted Jet Differentials via Slanted Vector Fields

To construct an additional family of independent negatively twisted jet differentials, we employ Siu's method of slanted vector fields [65]. This approach generalizes to k-jet spaces the original technique developed for 1-jet spaces by Clemens [11], Ein [24], and Voisin [70] in their study of rational curves.

The method was first implemented by Păun [54] for compact surfaces in $\mathbb{P}^3$. Rousseau [58] subsequently established the logarithmic theory for pairs $(\mathbb{P}^2, D)$ with curve configurations D, while in [57, 56] he further extended both Păun's setting and the logarithmic case to threefolds. For hypersurfaces $H \subset \mathbb{P}^n$ of arbitrary dimension, Merker [43] established the complete generalization, thereby realizing Siu's original vision concerning generic global generation. The logarithmic counterpart for pairs $(\mathbb{P}^n, H)$ was subsequently obtained by Darondeau [15].

This section adapts the standard technique of Siu's slanted vector fields to the case of three generic conics in $\mathbb{P}^2$. The content is entirely classical and no originality is claimed.

5.1. Logarithmic Slanted Vector Fields for Three Conics in $\mathbb{P}^2$.

The framework presented in this subsection specializes Rousseau's logarithmic setting in $\mathbb{P}^2$ to the configurations of three conics. We emphasize that this subsection contains no original contributions and is included solely for completeness and the reader's convenience.

Let

$$\mathscr{X} = \cup_{\lambda=1}^3 \mathscr{X}_\lambda \subset \mathbb{P}^2 \times \textstyle\prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}} \tag{5.1}$$

denote the universal family of algebraic curves in $\mathbb{P}^2$ consisting of three components of degrees d_1, d_2, and d_3, respectively. Here, each $\mathscr{X}_\lambda$ is the universal curve of degree d_λ, defined by the equation $\sum_{|\alpha|=d_\lambda} A^\lambda_\alpha Z^\alpha = 0$, where $[A^\lambda] = [A^\lambda_\alpha]_{|\alpha|=d_\lambda} \in \mathbb{P}^{N_{d_\lambda}}$ are the homogeneous coefficients and $[Z] = [Z_0 : Z_1 : Z_2] \in \mathbb{P}^2$ are the projective coordinates. For a multi-index $\alpha = (\alpha_0, \alpha_1, \alpha_2) \in \mathbb{N}^3$ with length $|\alpha| := \alpha_0 + \alpha_1 + \alpha_2$, we employ the notation $Z^\alpha := Z_0^{\alpha_0} Z_1^{\alpha_1} Z_2^{\alpha_2}$. The dimension of the parameter space for each component is given by $N_{d_\lambda} = \binom{d_\lambda+2}{2} - 1$.

For any integer $k \geqslant 1$, the *vertical logarithmic k-jet space* is then defined as

$$\overline{J^v_k}\left(\mathbb{P}^2 \times \textstyle\prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}}\right) := \ker (\mathrm{pr}_2)_* \subset J_k\left(\mathbb{P}^2 \times \textstyle\prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}}, -\log \mathscr{X}\right),$$

where $\mathrm{pr}_2 \colon \mathbb{P}^2 \times \prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}} \longrightarrow \prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}}$ denotes the natural projection onto the parameter space and $J_k\big(\mathbb{P}^2 \times \prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}}, -\log \mathscr{X}\big)$ is the logarithmic k-jet bundle along the universal family $\mathscr{X}$.

We define the universal intersection family

$$\mathscr{Y} = \cap_{\lambda=1}^3 \mathscr{Y}_\lambda \subset \mathbb{P}^5 \times \textstyle\prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}+1},$$

where for each $1 \leqslant \lambda \leqslant 3$, the hypersurface $\mathscr{Y}_\lambda$ is the universal family of degree d_λ hypersurfaces in $\mathbb{P}^5$, given explicitly by $\mathscr{Y}_\lambda := \left\{ A^\lambda_{\mathbf{0}} Z_{2+\lambda}^{d_\lambda} + \sum_{|\alpha|=d_\lambda} A^\lambda_\alpha Z^\alpha = 0 \right\} \subset \mathbb{P}^5 \times \mathbb{P}^{N_{d_\lambda}+1}$. Here, the space $\mathbb{P}^5$ has homogeneous coordinates $[Z : Z_3 : Z_4 : Z_5] = [Z_0 : \cdots : Z_5]$, and the parameter space $\mathbb{P}^{N_{d_\lambda}+1}$ is equipped with homogeneous coordinates $[A^\lambda : A^\lambda_{\mathbf{0}}] = [\ldots : A^\lambda_\alpha : \ldots : A^\lambda_{\mathbf{0}}]_{|\alpha|=d_\lambda}$.

For notational convenience, we denote the exceptional locus $\{Z = \mathbf{0}\} \cup \cup_{\lambda=1}^3 \{A^\lambda = \mathbf{0}\}$, where $\{Z = \mathbf{0}\}$ denotes the subset of $\mathbb{P}^5$ defined by $Z_0 = Z_1 = Z_2 = 0$ and $\{A^\lambda = \mathbf{0}\}$ refers to the locus in $\mathbb{P}^{N_{d_\lambda}+1}$ where all coordinates vanish except $A^\lambda_{\mathbf{0}}$. There exists a natural forgetful map

$$\pi \colon \mathbb{P}^5 \times \textstyle\prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}+1} \setminus \left(\{Z = \mathbf{0}\} \cup \cup_{\lambda=1}^3 \{A^\lambda = \mathbf{0}\} \right) \longrightarrow \mathbb{P}^2 \times \textstyle\prod_{\lambda=1}^3 \mathbb{P}^{N_{d_\lambda}}$$

that forgets the auxiliary coordinates. Explicitly, π is given by

$$\left([Z:Z_3:Z_4:Z_5],[A^1:A^1_{\mathbf{0}}],[A^2:A^2_{\mathbf{0}}],[A^3:A^3_{\mathbf{0}}]\right)\longmapsto\left([Z],[A^1],[A^2],[A^3]\right).$$

We observe the containment relation

$$\mathscr{Y}\cap\left(\{Z=\mathbf{0}\}\cup\cup_{\lambda=1}^{3}\{A^\lambda=\mathbf{0}\}\right)\subset\mathscr{Y}\cap\left(\cup_{\lambda=1}^{3}\left(\{A^\lambda_{\mathbf{0}}=0\}\cup\{A^\lambda=\mathbf{0}\}\right)\right),\tag{5.2}$$

from the inclusion $\mathscr{Y}\cap\{Z=\mathbf{0}\}\subset\mathscr{Y}\cap\cup_{\lambda=1}^{3}\{A^\lambda_{\mathbf{0}}=0\}$, which is deduced by the definition of $\mathscr{Y}$.

Now we consider $\mathscr{Y}^*:=\mathscr{Y}\setminus\left(\cup_{\lambda=1}^{3}\left(\{A^\lambda_{\mathbf{0}}=0\}\cup\{A^\lambda=\mathbf{0}\}\right)\right)$ on which the restricted forgetful map

$$\pi|_{\mathscr{Y}^*}\colon\mathscr{Y}^*\to\mathbb{P}^2\times\textstyle\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}}$$

is well-defined according to the containment relation (5.2). Furthermore, we obtain the identification of the preimage $(\pi|_{\mathscr{Y}^*})^{-1}(\mathscr{X})=\{Z_3Z_4Z_5=0\}=:\mathscr{D}$. This shows that π induces a logarithmic morphism from $(\mathscr{Y}^*,\mathscr{D})$ to $\left(\mathbb{P}^2\times\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}},\mathscr{X}\right)$, which consequently gives rise to a dominant map between jet spaces:

$$\pi_{[2]}\colon\overline{J^v_2}(\mathscr{Y}^*)\longrightarrow\overline{J^v_2}\left(\mathbb{P}^2\times\textstyle\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}}\right).\tag{5.3}$$

Here, $\overline{J^v_2}(\mathscr{Y}^*)$ represents the space of vertical logarithmic 2-jets with respect to the normal crossing divisor $\mathscr{D}=\{Z_3Z_4Z_5=0\}$. This construction allows us to perform all local computations on the more tractable space $\mathscr{Y}^*$ through pullback via π, where we have access to explicit coordinate expressions.

On the affine chart $\{Z_0\neq0\}\times\prod_{\lambda=1}^{3}\{A^\lambda_{\mathbf{0}}\neq0\}$, we employ the affine coordinates $z_i:=Z_i/Z_0$ for $i=1,\dots,5$ and $a^\lambda_\alpha:=A^\lambda_{\alpha_0\alpha_1\alpha_2}/A^\lambda_{\mathbf{0}}$ with $\alpha_0=d_\lambda-\alpha_1-\alpha_2$. To streamline notation, for each jet of order $j=1,2$ we define:

$$\xi^{(j)}_i:=\begin{cases}z^{(j)}_i & \text{for } i=1,2,\\ (\log z_{2+\lambda})^{(j)} & \text{for } i=3,4,5\ (\lambda=i-2).\end{cases}$$

Then local coordinates for $\overline{J_2}\left(\{Z_0\neq0\}\times\prod_{\lambda=1}^{3}\{A^\lambda_{\mathbf{0}}\neq0\}\right)\subset J_2\left(\mathbb{P}^5\times\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}+1},-\log\mathscr{D}\right)$ are given by the coordinate tuple $\left(z_i,a^\lambda_\alpha,\xi'_{i_1},\xi''_{i_2}\right)$.

Following the strategy of Siu [65] and the realizations of Păun [54] and Rousseau [58], we consider the following family of Siu's slanted vector fields with pole order 7:

Theorem 5.1. *Let Σ_0 be the zero locus of the 2×2 Wronskian $\det(\xi^{(j)}_i)_{1\leqslant i,j\leqslant2}$:*

$$\Sigma_0=\left\{\left(z_i,a^\lambda_\alpha,\xi'_{i_1},\xi''_{i_2}\right)\,\middle|\,\det\left(\xi^{(j)}_i\right)_{1\leqslant i,j\leqslant2}=0\right\}\subset\overline{J^v_2}\left(\mathscr{Y}\cap\left(\{Z_0\neq0\}\times\textstyle\prod_{\lambda=1}^{3}\{A^\lambda_{\mathbf{0}}\neq0\}\right)\right),$$

and Σ be the closure of Σ_0 in $\overline{J^v_2}(\mathscr{Y}^*)$. Then for some fixed $q_1,q_2,q_3\in\mathbb{Z}_{>0}$, the twisted tangent bundle

$$T_{\overline{J^v_2}\left(\mathbb{P}^2\times\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}}\right)}\otimes p^*\mathrm{pr}_1^*\mathscr{O}_{\mathbb{P}^2}(7)\otimes p^*\mathrm{pr}_2^*\mathscr{O}_{\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}}}(q_1,q_2,q_3),\tag{5.4}$$

is generated by its global sections at every point of $\overline{J^v_2}(\mathbb{P}^2\times\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}})\setminus\left(\pi_{[2]}(\Sigma)\cup p^{-1}(\mathscr{X})\right)$, where $\pi_{[2]}$ is introduced in (5.3), p is the natural projection

$$p\colon\overline{J^v_2}\left(\mathbb{P}^2\times\textstyle\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}}\right)\longrightarrow\mathbb{P}^2\times\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}},$$

and pr_1 (resp. pr_2) is the projection from $\mathbb{P}^2\times\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}}$ to $\mathbb{P}^2$ (resp. $\prod_{\lambda=1}^{3}\mathbb{P}^{N_{d_\lambda}}$). □

Remark 5.2. By the construction and proof of [58, Corollary 8 and Proposition 10], together with [43, Section 4.3], the vector fields in Theorem 5.1 are $\mathbb{G}_2$-invariant (see (2.3)). Consequently, their jet-direction coefficients carry exactly the weights compensating those lost under differentiation, and hence their action preserves the weighted degree in the jet variables.

Moreover, as in [58, Remark 10], if the image of the 2-jet lift $j_2(f)$ is contained in $\pi_{[2]}(\Sigma)$, then the image of f is contained in a line and hence is algebraically degenerate. Therefore, for an algebraically nondegenerate entire curve f, its 2-jet lift is not entirely contained in $\pi_{[2]}(\Sigma)$, and it suffices to work on the nonempty open set where the lifted jet lies outside $\pi_{[2]}(\Sigma)$.

5.2. **Producing an Independent 2-Jet via Siu's Slanted Vector Fields of Pole Order 7.**

Proposition 5.3 (The second global jet differential)**.** *Let $P := \{\omega_{1,a}\}_{a\in U}$ be the holomorphic family supplied by Proposition 4.3, with weighted degree m and negative base twist $t \geqslant 8$. For a generic parameter a, one can choose a pole-order-7 slanted vector field V such that*

$$\omega_{2,a} := \mathscr{L}_V P|_a \in H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}_a)\otimes \mathscr{O}_{\mathbb{P}^2}(7-t)\big)$$

is nonzero and the common zero locus of $\omega_{1,a}$ and $\omega_{2,a}$ in $X_{2,a}$ has codimension at least 2 modulo $\Gamma_{2,a}$. Consequently, if $f^*\omega_{1,a}\equiv 0$, then either

$$T_f(r) \leqslant \frac{m}{t-7}\sum_{i=1}^{3} N_f^{[1]}(r,\mathscr{C}_i) + O\big(\log T_f(r)+\log r\big) \quad \|,$$

or $f_{[1]}(\mathbb{C})$ is algebraically degenerate in $X_{1,a}$.

This is Siu's global differentiation mechanism [65], in the logarithmic form developed by Rousseau [58]; see also the explicit invariant formulation in [19, pp. 171–173]. On the universal vertical jet space, P is locally expressed as an equivariant polynomial function of the jet variables, and a slanted vector field acts on it naturally by derivation. Equivariance makes the result independent of the homogeneous chart and preserves the weighted degree and reparameterization invariance.

Proof. Regard the family $\{\omega_{1,a}\}_{a\in U}$ as the equivariant holomorphic map

$$P : \overline{J}^v_2\Big(\mathbb{P}^2\times\textstyle\prod_{\lambda=1}^3\mathbb{P}^{N_{d_\lambda}}\Big)\big|_{\mathrm{pr}_2^{-1}(U)} \longrightarrow p^*\mathrm{pr}_1^*\mathscr{O}_{\mathbb{P}^2}(-t).$$

If V is a global invariant section of the twisted tangent bundle (5.4), its natural action on equivariant functions gives

$$\mathscr{L}_V P : \overline{J}^v_2\Big(\mathbb{P}^2\times\textstyle\prod_{\lambda=1}^3\mathbb{P}^{N_{d_\lambda}}\Big)\big|_{\mathrm{pr}_2^{-1}(U)} \longrightarrow p^*\mathrm{pr}_1^*\mathscr{O}_{\mathbb{P}^2}(7-t),$$

where the parameter-space twist becomes a constant one-dimensional factor after restriction to a fixed fibre. Since V is invariant, $\mathscr{L}_V P$ is again an invariant polynomial of weighted degree m in the jet variables. Its restriction to the fibre over a is therefore the asserted global differential $\omega_{2,a}$.

By Proposition 4.3, the residual zero divisor of $\omega_{1,a}$ is irreducible and reduced modulo $\Gamma_{2,a}$. The global generation in Theorem 5.1 allows us to choose V non-tangent to that divisor at a general smooth point. Hence $\mathscr{L}_V P$ does not vanish identically there, and the common zero locus has codimension at least 2 modulo $\Gamma_{2,a}$.

If $f^*\omega_{2,a}\not\equiv 0$, apply Theorem 1.2 to $\omega_{2,a}$, whose negative twist is $t-7$. If both pullbacks vanish, then $f_{[2]}(\mathbb{C})$ lies in their common zero locus. Its image in $X_{1,a}$ has dimension at most 2, strictly smaller than $\dim X_{1,a}=3$; hence $f_{[1]}$ is algebraically degenerate. □

6. Refinement of Demailly–El Goul's Arguments

6.1. **More Riemann-Roch Computations.** In Proposition 4.3, we show the existence of a nonzero negatively twisted invariant logarithmic 2-jet differential

$$\omega_1 \in H^0\big(X_2, \mathscr{O}_{X_2}(m)\otimes \pi_{2,0}^*\mathscr{O}_X(-t)\big),$$

for some positive integers $m,t\geqslant 1$ with $t/m\geqslant 1/5$, whose zero divisor on the second level of the Demailly-Semple tower X_2 is of the form $Z+b_1\Gamma_2$, where $Z\subset X_2$ is irreducible and reduced. Moreover, we have the linear equivalence

$$Z \sim b_1u_1+b_2u_2-t\pi_{2,0}^*h, \quad (b_1,b_2)\in\mathbb{Z}^2, \quad b_1\geqslant 2b_2>0, \quad b_1+b_2=m, \quad t\in\mathbb{Z}_+, \tag{6.1}$$

where $u_1=\pi_{2,1}^*\mathscr{O}_{X_1}(1)$, $u_2=\mathscr{O}_{X_2}(1)$, and $h=\mathscr{O}_{\mathbb{P}^2}(1)$.

Hereafter, we use the standard notation:

$$\mathscr{O}_{X_2}(b_1,b_2) := \pi_{2,1}^*\mathscr{O}_{X_1}(b_1)\otimes\mathscr{O}_{X_2}(b_2), \quad b_1,b_2\in\mathbb{Z}.$$

Using the same argument as in Demailly–El Goul [18, Proposition 3.4] — Miyaoka's method [45]; see also Lu–Yau [40] — one can show that $\mathscr{O}_{X_2}(2,1)|_Z$ is big, provided that $t<m$. Now, our goal is to determine an effective small rational number $\tau>0$ such that the $\mathbb{Q}$-line bundle $\big(\mathscr{O}_{X_2}(2,1)\otimes\pi_{2,0}^*\mathscr{O}_{\mathbb{P}^2}(-\tau)\big)|_Z$ remains big. This will be achieved through Riemann-Roch computations and the following proposition observed by Demailly [17, p. 73].

Proposition 6.1. *Let $a_1,a_2\in\mathbb{Z}$.*

(1) *If* $a_2 = -1$, *then*

$$R^q(\pi_{2,0})_*\mathscr{O}_{X_2}(a_1,-1) = 0 \qquad \text{for every } q \geqslant 0.$$

(2) *If* $a_2 \geqslant 0$, *then*

$$R^q(\pi_{2,0})_*\mathscr{O}_{X_2}(a_1,a_2) = 0 \qquad \text{for every } q \geqslant 2,$$

and

$$R^1(\pi_{2,0})_*\mathscr{O}_{X_2}(a_1,a_2) = 0 \quad \Longleftrightarrow \quad a_1 - 2a_2 \geqslant -1.$$

In particular, for every $p \geqslant 0$,

$$R^q(\pi_{2,0})_*\mathscr{O}_{X_2}(2p,p) = 0 \qquad \text{for every } q \geqslant 1.$$

Proof. Set

$$\pi_{2,1}\colon X_2 = \mathbb{P}(V_1) \longrightarrow X_1, \qquad \pi_{1,0}\colon X_1 \longrightarrow X_0.$$

Recall that V_1 is the rank-two directed bundle defined by (2.7), and that V_1^* denotes its dual. We use the convention that $\mathbb{P}(V_1)$ parametrizes lines in the fibers of V_1, so that $\mathscr{O}_{X_2}(-1) \subset \pi_{2,1}^* V_1$ is the tautological line subbundle.

Suppose first that $a_2 = -1$. The restriction of $\mathscr{O}_{X_2}(a_1,-1)$ to every fiber of $\pi_{2,1}$ is $\mathscr{O}_{\mathbb{P}^1}(-1)$, whose cohomology vanishes in every degree. Cohomology and base change therefore gives

$$R^q(\pi_{2,1})_*\mathscr{O}_{X_2}(a_1,-1) = 0 \qquad (q \geqslant 0),$$

and the first assertion follows from $\pi_{2,0} = \pi_{1,0} \circ \pi_{2,1}$.

Assume now that $a_2 \geqslant 0$. By the projective bundle formula and the projection formula,

$$R^q(\pi_{2,1})_*\mathscr{O}_{X_2}(a_1,a_2) = 0 \quad (q \geqslant 1), \qquad (\pi_{2,1})_*\mathscr{O}_{X_2}(a_1,a_2) = \mathscr{O}_{X_1}(a_1) \otimes S^{a_2} V_1^*. \tag{6.2}$$

Hence the Leray spectral sequence yields

$$R^q(\pi_{2,0})_*\mathscr{O}_{X_2}(a_1,a_2) \simeq R^q(\pi_{1,0})_*\big(\mathscr{O}_{X_1}(a_1) \otimes S^{a_2} V_1^*\big).$$

Fix $x \in X_0$ and set

$$F_x := \pi_{1,0}^{-1}(x) \simeq \mathbb{P}^1.$$

Restricting (2.7) to F_x gives

$$0 \longrightarrow \mathscr{O}_{\mathbb{P}^1}(2) \longrightarrow V_1|_{F_x} \longrightarrow \mathscr{O}_{\mathbb{P}^1}(-1) \longrightarrow 0.$$

This sequence splits, since

$$\operatorname{Ext}^1\big(\mathscr{O}_{\mathbb{P}^1}(-1), \mathscr{O}_{\mathbb{P}^1}(2)\big) = H^1\big(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(3)\big) = 0.$$

Thus

$$V_1^*|_{F_x} \simeq \mathscr{O}_{\mathbb{P}^1}(-2) \oplus \mathscr{O}_{\mathbb{P}^1}(1),$$

and, since $\mathscr{O}_{X_1}(1)|_{F_x} \simeq \mathscr{O}_{\mathbb{P}^1}(1)$,

$$\big(\mathscr{O}_{X_1}(a_1) \otimes S^{a_2} V_1^*\big)\big|_{F_x} \simeq \bigoplus_{j=0}^{a_2} \mathscr{O}_{\mathbb{P}^1}(a_1 + a_2 - 3j). \tag{6.3}$$

The smallest degree in (6.3) is $a_1 - 2a_2$. Since

$$H^1\big(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(d)\big) = 0 \quad \Longleftrightarrow \quad d \geqslant -1,$$

the fiberwise first cohomology vanishes precisely when $a_1 - 2a_2 \geqslant -1$. The splitting type is independent of x, so Grauert's theorem gives

$$R^1(\pi_{2,0})_*\mathscr{O}_{X_2}(a_1,a_2) = 0 \quad \Longleftrightarrow \quad a_1 - 2a_2 \geqslant -1.$$

Finally, $\pi_{1,0}$ has one-dimensional fibers; hence

$$R^q(\pi_{1,0})_*\big(\mathscr{O}_{X_1}(a_1) \otimes S^{a_2} V_1^*\big) = 0 \qquad (q \geqslant 2).$$

Together with (6.2), this proves the remaining assertion. Taking $(a_1,a_2) = (2p,p)$ gives $a_1 - 2a_2 = 0$, and therefore all higher direct images vanish. □

Proposition 6.2. *If* $0 < \frac{t}{m} < 1$ *and* $0 \leqslant \tau < \tau_1(\frac{t}{m}), \tau \in \mathbb{Q}$, *where*

$$\tau_1\Big(\frac{t}{m}\Big) = \frac{3}{2}\Big(4 - \frac{t}{m}\Big) - \frac{\sqrt{3}}{2}\sqrt{32 - 8\frac{t}{m} + 3\Big(\frac{t}{m}\Big)^2},$$

then the restricted $\mathbb{Q}$*-line bundle* $\big(\mathscr{O}_{X_2}(2,1) \otimes \pi_{2,0}^* \mathscr{O}_{\mathbb{P}^2}(-\tau)\big)|_Z$ *is big on* Z.

Proof. We begin by recalling some key intersection formulas on the Demailly–Semple tower. By [25, p. 478], we have the identities:

$$u_1^2+\bar{c}_1u_1+\bar{c}_2=0 \quad \text{and} \quad u_2^2+\bar{c}_1(V_1)u_2+\bar{c}_2(V_1)=0,$$

where $\bar{c}_1(V_1)=\bar{c}_1+u_1$ (see (2.6)) and $\bar{c}_2(V_1)=\bar{c}_2-u_1^2=2\bar{c}_2+\bar{c}_1u_1$.

On the second level X_2 of the Demailly-Semple tower, the following identities hold:

$$u_1^4=0, \quad u_1^3u_2=\bar{c}_1^2-\bar{c}_2, \quad u_1^2u_2^2=\bar{c}_2, \quad u_1u_2^3=\bar{c}_1^2-3\bar{c}_2, \quad u_2^4=5\bar{c}_2-\bar{c}_1^2.$$

Furthermore, for any pullback $F\in\pi_{2,0}^*\mathrm{Pic}(\mathbb{P}^2)$, we have:

$$u_1^3\cdot F=0, \quad u_1^2u_2\cdot F=-\bar{c}_1\cdot F, \quad u_1u_2^2\cdot F=0, \quad u_2^3\cdot F=0.$$

A crucial part of our approach relies on the following additional identities:

$$u_1^2\cdot F_1\cdot F_2=0, \quad u_1u_2\cdot F_1\cdot F_2=F_1\cdot F_2, \quad u_2^2\cdot F_1\cdot F_2=-F_1\cdot F_2, \qquad \forall F_1,F_2\in\pi_{2,0}^*\mathrm{Pic}(\mathbb{P}^2).$$

Then by direct computation, we can obtain

$$\left(2u_1+u_2-\tau\pi_{2,0}^*h\right)^3\cdot Z=m\left(13\bar{c}_1^2-9\bar{c}_2\right)+12t(\bar{c}_1\cdot h)+12m(\bar{c}_1\cdot h)\tau+3m\left(h^2\right)\tau^2+9\left(h^2\right)t\tau. \tag{6.4}$$

In our setting of three generic conics, we have $\bar{c}_1=-3h$ and $\bar{c}_2=9h^2$. Substituting these into (6.4), together with $h^2=1$, yields:

$$\left(2u_1+u_2-\tau\pi_{2,0}^*h\right)^3\cdot Z=3\left(m\tau^2-3(4m-t)\tau+12(m-t)\right). \tag{6.5}$$

This quadratic polynomial has two real roots:

$$\tau_1\left(\tfrac{t}{m}\right)=\frac{3}{2}\left(4-\frac{t}{m}\right)-\frac{\sqrt{3}}{2}\sqrt{32-8\frac{t}{m}+3\left(\frac{t}{m}\right)^2}<\tau_2\left(\frac{t}{m}\right)=\frac{3}{2}\left(4-\frac{t}{m}\right)+\frac{\sqrt{3}}{2}\sqrt{32-8\frac{t}{m}+3\left(\frac{t}{m}\right)^2}, \tag{6.6}$$

because $32-8\frac{t}{m}+3(\frac{t}{m})^2>0$. Moreover, $\tau_1(\frac{t}{m})>0$ if and only if $\frac{t}{m}<1$.

Thus when $0<\frac{t}{m}<1$ and $0\leqslant\tau<\tau_1(\frac{t}{m})$, we have $(2u_1+u_2-\tau\pi_{2,0}^*h)^3\cdot Z>0$. Riemann-Roch Theorem then gives, for sufficiently divisible integer $p\gg1$, that:

$$\begin{aligned}&h^0\left(Z,(\mathscr{O}_{X_2}(2,1)\otimes\pi_{2,0}^*\mathscr{O}_{\mathbb{P}^2}(-\tau))^p|_Z\right)+h^2\left(Z,(\mathscr{O}_{X_2}(2,1)\otimes\pi_{2,0}^*\mathscr{O}_{\mathbb{P}^2}(-\tau))^p|_Z\right)\\ \geqslant&\chi\left(Z,(\mathscr{O}_{X_2}(2,1)\otimes\pi_{2,0}^*\mathscr{O}_{\mathbb{P}^2}(-\tau))^p|_Z\right)=\frac{\left(2u_1+u_2-\tau\pi_{2,0}^*h\right)^3\cdot Z}{3!}p^3+O(p^2)\gg1.\end{aligned}$$

Now we prove that

$$H^2\left(Z,\left(\mathscr{O}_{X_2}(2,1)\otimes\pi_{2,0}^*\mathscr{O}_{\mathbb{P}^2}(-\tau)\right)^p|_Z\right)=0 \tag{6.7}$$

for all sufficiently divisible $p\gg1$. Put

$$\mathscr{L}_p=\mathscr{O}_{X_2}(2p,p)\otimes\pi_{2,0}^*\mathscr{O}_{\mathbb{P}^2}(-\tau p).$$

Tensoring $0\to\mathscr{O}_{X_2}(-Z)\to\mathscr{O}_{X_2}\to\mathscr{O}_Z\to0$ with $\mathscr{L}_p$ gives

$$h^2(Z,\mathscr{L}_p|_Z)\leqslant h^2(X_2,\mathscr{L}_p)+h^3(X_2,\mathscr{L}_p\otimes\mathscr{O}_{X_2}(-Z)). \tag{6.8}$$

For the first term, Proposition 6.1 applies because $2p-2p=0\geqslant-1$. Hence

$$R^q(\pi_{2,0})_*\mathscr{O}_{X_2}(2p,p)=0 \qquad (q\geqslant1)$$

and

$$(\pi_{2,0})_*\mathscr{O}_{X_2}(2p,p)\simeq E_{2,3p}T^*_{\mathbb{P}^2}(\log\mathscr{C}).$$

Indeed, in filtration layer q a weighted-degree $3p$ invariant monomial has $3p-3q$ first-derivative factors and q Wronskian factors. On X_2 its class is

$$\mathscr{O}_{X_2}(3p-q,q)=\mathscr{O}_{X_2}(3p)\left(-(3p-q)\Gamma_2\right).$$

Since $0\leqslant q\leqslant p$, every section of $\mathscr{O}_{X_2}(3p)$ coming from an invariant differential vanishes along Γ_2 at least to order $2p$. Hence

$$(\pi_{2,0})_*\mathscr{O}_{X_2}(2p,p)\simeq(\pi_{2,0})_*\mathscr{O}_{X_2}(3p)\simeq E_{2,3p}T^*_{\mathbb{P}^2}(\log\mathscr{C}),$$

which justifies the displayed zeroth-direct-image identification. By Leray and the projection formula,

$$H^2(X_2,\mathscr{L}_p)\simeq H^2\left(\mathbb{P}^2,E_{2,3p}T^*_{\mathbb{P}^2}(\log\mathscr{C})\otimes\mathscr{O}_{\mathbb{P}^2}(-\tau p)\right).$$

The argument of Proposition 4.2, with jet degree $3p$ and $\delta = \tau/9$, gives the vanishing of this group. Indeed,

$$0 \leqslant \tau < \tau_1(t/m) \leqslant 6 - 2\sqrt{6} < 3,$$

so that $0 \leqslant \delta < 1/3$. Therefore

$$H^2(X_2, \mathscr{L}_p) = 0. \tag{6.9}$$

It remains to treat the second term in (6.8). By (6.1), set

$$\mathscr{M}_p := \mathscr{L}_p \otimes \mathscr{O}_{X_2}(-Z) = \mathscr{O}_{X_2}(2p - b_1, p - b_2) \otimes \pi_{2,0}^* \mathscr{O}_{\mathbb{P}^2}(t - \tau p).$$

Write

$$r := b_1 - 2b_2 \geqslant 0.$$

Choose p so large that $p - b_2 \geqslant 0$ and, when $r \geqslant 2$,

$$p - b_2 \geqslant \left\lfloor \frac{r-2}{3} \right\rfloor.$$

Since $p - b_2 \geqslant 0$,

$$R^q(\pi_{2,1})_* \mathscr{M}_p = 0 \quad (q \geqslant 1),$$

whereas $(\pi_{2,1})_* \mathscr{M}_p$ has a filtration induced by

$$0 \longrightarrow \mathscr{O}_{X_1}(1) \longrightarrow V_1^* \longrightarrow \pi_{1,0}^* \overline{K}_{\mathbb{P}^2} \otimes \mathscr{O}_{X_1}(-2) \longrightarrow 0.$$

After reindexing by $j = p - b_2 - k$, the corresponding graded terms are

$$\mathscr{Q}_{p,k} = \pi_{1,0}^* \Big(\overline{K}_{\mathbb{P}^2}^{\,p-b_2-k} \otimes \mathscr{O}_{\mathbb{P}^2}(t - \tau p) \Big) \otimes \mathscr{O}_{X_1}(-r + 3k), \qquad 0 \leqslant k \leqslant p - b_2. \tag{6.10}$$

Set

$$A_{p,k} := \overline{K}_{\mathbb{P}^2}^{\,p-b_2-k} \otimes \mathscr{O}_{\mathbb{P}^2}(t - \tau p), \qquad d_k := -r + 3k,$$

so that

$$\mathscr{Q}_{p,k} = \pi_{1,0}^* A_{p,k} \otimes \mathscr{O}_{X_1}(d_k).$$

If $d_k \geqslant -1$, then the projective bundle formula and the projection formula give

$$R^1(\pi_{1,0})_* \mathscr{Q}_{p,k} \simeq A_{p,k} \otimes R^1(\pi_{1,0})_* \mathscr{O}_{X_1}(d_k) = 0.$$

Since $\pi_{1,0}$ has one-dimensional fibers and $\dim \mathbb{P}^2 = 2$, the Leray spectral sequence yields

$$H^3(X_1, \mathscr{Q}_{p,k}) \simeq H^2\big(\mathbb{P}^2, R^1(\pi_{1,0})_* \mathscr{Q}_{p,k}\big) = 0.$$

Consequently, only the indices satisfying $d_k \leqslant -2$ may contribute; equivalently,

$$0 \leqslant k \leqslant \left\lfloor \frac{r-2}{3} \right\rfloor.$$

The only possible nonvanishing terms are therefore indexed by the fixed finite set

$$0 \leqslant k \leqslant \left\lfloor \frac{r-2}{3} \right\rfloor. \tag{6.11}$$

For such k, relative Serre duality for $\pi_{1,0} \colon X_1 = \mathbb{P}(T_{\mathbb{P}^2}(-\log \mathscr{C})) \to \mathbb{P}^2$ gives

$$R^1(\pi_{1,0})_* \mathscr{Q}_{p,k} \simeq S^{r-3k-2} T_{\mathbb{P}^2}(-\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}\big((3 - \tau)p + t - 3(b_2 + k + 1)\big), \tag{6.12}$$

because $\overline{K}_{\mathbb{P}^2} \simeq \mathscr{O}_{\mathbb{P}^2}(3)$. For each k in (6.11), the symmetric power in (6.12) is independent of p, while the twist tends to $+\infty$ since $\tau < 3$. Serre vanishing therefore gives

$$H^2\big(\mathbb{P}^2, R^1(\pi_{1,0})_* \mathscr{Q}_{p,k}\big) = 0 \qquad (p \gg 1).$$

The Leray spectral sequence for $\pi_{1,0}$ now yields $H^3(X_1, \mathscr{Q}_{p,k}) = 0$ for every graded term of the filtration. Induction along the filtration gives

$$H^3(X_1, (\pi_{2,1})_* \mathscr{M}_p) = 0.$$

Using $R^q(\pi_{2,1})_* \mathscr{M}_p = 0$ once more, we obtain

$$H^3(X_2, \mathscr{M}_p) = 0. \tag{6.13}$$

Combining (6.8), (6.9), and (6.13) proves (6.7). Thus Riemann–Roch gives

$$h^0(Z, \mathscr{L}_p|_Z) \geqslant \frac{(2u_1 + u_2 - \tau \pi_{2,0}^* h)^3 \cdot Z}{3!}\, p^3 + O(p^2),$$

and the positive leading coefficient proves that the restricted $\mathbb{Q}$-line bundle is big. □

6.2. **A Refined Estimate.** Here is a key step in proving our main theorem.

Lemma 6.3 (Tautological evaluation of a restricted section)**.** *Let $Z \subset X_2$ be an effective Cartier divisor, let $p, \widetilde{t} \in \mathbb{Z}_{>0}$, and let*

$$\sigma \in H^0\big(Z, \mathscr{O}_{X_2}(2p,p) \otimes \pi_{2,0}^* \mathscr{O}_{\mathbb{P}^2}(-\widetilde{t})\big|_Z\big)\,.$$

Suppose that $g\colon \mathbb{C} \to \mathbb{P}^2$ is nonconstant, $g(\mathbb{C}) \not\subset \mathscr{C}$, and that its logarithmic Semple lift extends across its isolated indeterminacy points and satisfies $g_{[2]}(\mathbb{C}) \subset Z$. On $\mathbb{C} \setminus g^{-1}(\mathscr{C})$, regard $dg_{[1]}\colon T_{\mathbb{C}} \to g_{[1]}^* V_1$ as a differential in the directed structure. Wherever it is nonzero, its image is the tautological line $g_{[2]}^* \mathscr{O}_{X_2}(-1)$. It therefore defines, by meromorphic continuation across the remaining discrete set, the intrinsic tautological derivative

$$\mathscr{D}g_{[1]} \in H^0_{\mathrm{mer}}\Big(\mathbb{C}, K_{\mathbb{C}} \otimes g_{[2]}^* \mathscr{O}_{X_2}(-1)\Big)\,.$$

Here $H^0_{\mathrm{mer}}(Y,L)$ denotes the vector space of global meromorphic sections of a line bundle L on a complex manifold Y. The canonical inclusion

$$\iota_p\colon \mathscr{O}_{X_2}(2p,p) = \mathscr{O}_{X_2}(3p)(-2p\Gamma_2) \hookrightarrow \mathscr{O}_{X_2}(3p)$$

defines an intrinsic meromorphic section

$$s_{\sigma,g} := \Big\langle g_{[2]}^*(\iota_p\sigma), (\mathscr{D}g_{[1]})^{\otimes 3p} \Big\rangle \in H^0_{\mathrm{mer}}\Big(\mathbb{C}, K_{\mathbb{C}}^{\otimes 3p} \otimes g^* \mathscr{O}_{\mathbb{P}^2}(-\widetilde{t})\Big)\,, \tag{6.14}$$

This construction is independent of every choice of an ambient local lifting of σ. If $s_{\sigma,g} \not\equiv 0$, then its pole divisor satisfies

$$\mathrm{div}_\infty(s_{\sigma,g}) \leqslant 3p \sum_{i=1}^{3} (g^*\mathscr{C}_i)_{\mathrm{red}}, \tag{6.15}$$

and, outside a set of finite Lebesgue measure,

$$\int_0^{2\pi} \log^+ \|s_{\sigma,g}(re^{\sqrt{-1}\theta})\| \, \frac{d\theta}{2\pi} = O(\log T_g(r) + \log r)\,. \tag{6.16}$$

Proof. Since the map $g_{[2]}$ factors through Z, one may pull back $\iota_p\sigma$ directly from Z, without extending σ to a neighborhood. Pairing the two line bundles gives (6.14). If one nevertheless represents σ by ambient local liftings, two such liftings differ by a section in the ideal of Z; their pullbacks by $g_{[2]}$ therefore agree.

Equip the line bundles on X_2 and $\mathbb{P}^2$ with smooth Hermitian metrics. Since Z is projective, the norm of the holomorphic section $\iota_p\sigma$ is bounded on Z. Consequently, for a uniform constant $C > 0$,

$$\|s_{\sigma,g}\| \leqslant C\, \|\mathscr{D}g_{[1]}\|^{3p}.$$

This estimate uses only the intrinsic section on Z and does not extend it to a saturated inverse image in the Semple tower.

We next analyze the tautological derivative. Near a point of $g^{-1}(\mathscr{C})$, choose logarithmic coordinates with $\mathscr{C} = \{x_1 \cdots x_q = 0\}$, $q \leqslant 2$. The derivative $\mathrm{d}g_{[1]}$ has both base and fiber components. Its base components are those of g'; in the logarithmic boundary directions, they are $(x_i \circ g)'/(x_i \circ g)$. Since the extended first lift $g_{[1]}$ is holomorphic, its fiberwise projective-coordinate components are holomorphic. Thus $\mathscr{D}g_{[1]}$ has at most a simple pole along each $(g^*\mathscr{C}_i)_{\mathrm{red}}$. At an isolated zero of g' outside $g^{-1}(\mathscr{C})$, the regular Semple lift extends and the evaluation has only a removable singularity or a zero. If the critical point lies over $\mathscr{C}$, a logarithmic component such as $(x_i \circ g)'/(x_i \circ g)$ may still have a simple pole. It follows that $\mathscr{D}g_{[1]}$ has at most a simple pole along each $(g^*\mathscr{C}_i)_{\mathrm{red}}$. Taking its $3p$-th tensor power therefore gives (6.15).

Finally choose a finite logarithmic coordinate cover and, above it, a finite cover by relatively compact Semple charts. A partition of unity is used only to estimate the norm of the already-defined global section. On each chart, the base components of $\mathscr{D}g_{[1]}$ are logarithmic derivatives of local coordinate functions. A projective Semple coordinate u is a rational function on X_1 and, in a logarithmic frame, a ratio of two first logarithmic derivatives. Hence

$$T_{u \circ g_{[1]}}(r) = O\big(T_g(r) + \log r\big).$$

For a partition function χ supported in a relatively compact Semple chart, choose $a \notin \overline{u(\operatorname{supp}\chi)}$. On this support, $u-a$ is bounded above and away from zero, and

$$(u\circ g_{[1]})' = (u\circ g_{[1]}-a)\frac{(u\circ g_{[1]})'}{u\circ g_{[1]}-a}.$$

The logarithmic-derivative lemma applied to $u\circ g_{[1]}-a$ controls the vertical component by

$$O\big(\log^+ T_g(r)+\log r\big).$$

Summing this estimate and the analogous horizontal estimates over the finite cover gives (6.16); this is the partition-of-unity argument of [37, Thm. 3.1, (3.3)–(3.6)] with the preceding Semple-coordinate adaptation made explicit. □

Lemma 6.4 (Tautological evaluation of a restricted $\mathscr{O}_{X_2}(m)$-section)**.** *Let $Z\subset X_2$ be a projective subvariety and let*

$$\sigma \in H^0\big(Z, \mathscr{O}_{X_2}(m)\otimes \pi_{2,0}^*\mathscr{O}_{\mathbb{P}^2}(-s)|_Z\big), \qquad m,s\geqslant 1.$$

If $g_{[2]}(\mathbb{C})\subset Z$, the tautological pairing defines an intrinsic meromorphic section

$$\Big\langle g_{[2]}^*\sigma, (\mathscr{D}g_{[1]})^{\otimes m}\Big\rangle \in H^0_{\mathrm{mer}}\big(\mathbb{C}, K_{\mathbb{C}}^{\otimes m}\otimes g^*\mathscr{O}_{\mathbb{P}^2}(-s)\big).$$

If it is not identically zero, then

$$T_g(r)\leqslant \frac{m}{s}\sum_{i=1}^{3}N_g^{[1]}(r,\mathscr{C}_i)+O\big(\log T_g(r)+\log r\big) \quad \|.$$

Proof. The proof is the same intrinsic pairing and logarithmic/Semple-coordinate estimate as in Lemma 6.3, with $\mathscr{O}_{X_2}(m)$ paired directly against $\mathscr{O}_{X_2}(-m)$. Thus the reduced boundary pole divisor has coefficient at most m; Poincaré–Lelong and Jensen's formula give the stated inequality. □

Theorem 6.5. *Let $g\colon \mathbb{C}\to\mathbb{P}^2$ be an algebraically nondegenerate entire curve whose lift $g_{[2]}\colon \mathbb{C}\to X_2$ satisfies $g_{[2]}(\mathbb{C})\subset Z$. Suppose there exists a section*

$$\sigma \in H^0\Big(Z, \big(\mathscr{O}_{X_2}(2,1)^{\tilde m}\otimes\pi_{2,0}^*\mathscr{O}_X(-\tilde t)\big)|_Z\Big) \cong H^0\Big(Z, \big(\mathscr{O}_{X_2}(3)\otimes\mathscr{O}(-2\Gamma_2)\big)^{\tilde m}\otimes\pi_{2,0}^*\mathscr{O}_X(-\tilde t)\big|_Z\Big), \tag{6.17}$$

such that the canonical evaluation $s_{\sigma,g}$ of Lemma 6.3, with $p=\tilde m$, is not identically zero. Then the following Second Main Theorem type estimate holds:

$$T_g(r)\leqslant \frac{3\tilde m}{\tilde t}\sum_{i=1}^{3}N_g^{[1]}(r,\mathscr{C}_i)+o\big(T_g(r)\big) \quad \|.$$

The fixed double vertical factor is retained through the canonical inclusion in Lemma 6.3. The lemma, rather than an identification of an arbitrary local lifting with a jet polynomial, supplies the global meromorphic evaluation and the coefficient $3\tilde m/\tilde t$.

Proof. Equip $\mathscr{O}_X(-1)$ with the standard norm, whose Chern curvature is $-\omega_{\mathrm{FS}}$. By (6.17), the canonical evaluation $s_{\sigma,g}$ is a nonzero meromorphic section of $K_{\mathbb{C}}^{\otimes 3\tilde m}\otimes g^*\mathscr{O}_{\mathbb{P}^2}(-\tilde t)$. Give $K_{\mathbb{C}}$ the flat metric defined by dz. Poincaré–Lelong and Jensen's formula give

$$\tilde t\,T_g(r)\leqslant N(r,\operatorname{div}_\infty(s_{\sigma,g}))+\int_0^{2\pi}\log^+\|s_{\sigma,g}(re^{\sqrt{-1}\theta})\|\frac{d\theta}{2\pi}+O(1).$$

Applying (6.15) and (6.16), with $p=\tilde m$, yields

$$\tilde t\,T_g(r)\leqslant 3\tilde m\sum_{i=1}^{3}N_g^{[1]}(r,\mathscr{C}_i)+O(\log T_g(r)+\log r) \quad \|,$$

which is the asserted estimate. □

Lemma 6.6 (The zero-evaluation branch)**.** *In the setting of Theorem 6.5, assume that Z is irreducible and that $\sigma\neq 0$. If the canonical evaluation $s_{\sigma,g}$ is identically zero, then*

$$g_{[2]}(\mathbb{C})\subset Z(\sigma)\subsetneq Z,$$

and $g_{[1]}(\mathbb{C})$ is contained in a proper algebraic subset of X_1.

Proof. The tautological derivative $\mathscr{D}g_{[1]}$ is not identically zero, since otherwise $g_{[1]}$, and hence g, would be constant. On the dense open set where this derivative is nonzero and $g_{[2]}$ is outside Γ_2, the canonical inclusion ι_p is nonvanishing. The equality $s_{\sigma,g}=0$ therefore implies $\sigma(g_{[2]}(z))=0$. Analytic continuation gives the first inclusion. Since σ is a nonzero section on the irreducible threefold Z, $Z(\sigma)$ is proper and has dimension at most 2. Its image under the proper morphism $\pi_{2,1}$ is consequently a proper algebraic subset of the threefold X_1 containing $g_{[1]}(\mathbb{C})$. □

6.3. **An Effective Constant c in the Second Main Theorem for Three Conics.** Suppose there exists a nonzero negatively twisted invariant logarithmic 2-jet differential

$$\omega_1 \in H^0\big(\mathbb{P}^2, E_{2,m}T^*_{\mathbb{P}^2}(\log\mathscr{C})\otimes\mathscr{O}_{\mathbb{P}^2}(-t)\big)$$

satisfying $0<t<m$ and such that the divisor Z of ω_1 on the second level X_2 of the logarithmic Demailly-Semple tower for $(\mathbb{P}^2,\mathscr{C})$ is irreducible and reduced modulo Γ_2. Then, by the argument in Section 6.1, for any rational number $\tau\in(0,\tau_1(t/m))$, we obtain a nontrivial global section

$$\sigma\in H^0\left(Z,\big(\mathscr{O}_{X_2}(2,1)\otimes\pi_{2,0}^*\mathscr{O}_{\mathbb{P}^2}(-\tau)\big)^{\otimes\ell}\right)$$

for sufficiently divisible $\ell\gg 1$ (depending on τ).

Furthermore, if $f^*\omega_1\equiv 0$ and the canonical evaluation of σ along $f_{[2]}$ is nonzero, then Theorem 6.5 yields the estimate (1.7) with constant (see (6.6)) $c=\frac{3}{\tau}>\frac{3}{\tau_1(t/m)}$. If the canonical evaluation is zero, Lemma 6.6 shows that $f_{[1]}$ is algebraically degenerate, and Theorem 1.1 gives the stronger estimate $3T_f\leqslant\sum_i N_f^{[1]}+o(T_f)$. Thus, in either branch and by the arbitrariness of τ, the estimate (1.7) holds for any constant $c>3/\tau_1(\frac{t}{m})$.

Direct computation shows that $\inf_{0<x<1}\frac{3}{\tau_1(x)}=\frac{3+\sqrt{6}}{2}$. Thus $\frac{3}{\tau_1(t/m)}<c$ is equivalent to $c>\frac{3+\sqrt{6}}{2}$ and $0<\frac{t}{m}<\frac{4c^2-12c+3}{c(4c-3)}$.

Meanwhile, in **Case 1**, Proposition 5.3 gives the coefficient $m/(t-7)$. For this coefficient to be at most c, the positive integers m and t associated to the first family ω_1 must satisfy $t\geqslant\frac{m}{c}+7$.

Taking $c=19$, we obtain the desired estimate (1.7) for positive integer pairs (m,t) satisfying $0<t<\frac{1219}{1387}m$ or $t\geqslant\frac{m}{19}+7$. Note that $8\cdot\frac{1219}{1387}>7$, whereas $7\cdot\frac{1219}{1387}<7$. To exclude the exceptional cases of (m,t), it therefore suffices to prove Key Vanishing Lemma I. In fact, the exact real boundary is $c>(75+\sqrt{5241})/8$, so 19 is the smallest integral constant supplied by this argument.

Taking $c=5$, we obtain the desired estimate (1.7) for positive integer pairs (m,t) satisfying $0<t<\frac{43}{85}m$ or $t\geqslant\frac{m}{5}+7$. For $m\geqslant 23$ these two regions overlap, since $43m/85\geqslant m/5+7$. For $3\leqslant m\leqslant 22$, direct integer enumeration shows that, after using monotonicity in the negative twist, the only minimal uncovered pairs are precisely the eighteen pairs in Key Vanishing Lemma II. (For $m=20$ and $m=22$ there is no gap.)

Proof of Theorem 1.6. Work on the common nonempty Zariski-open locus on which Proposition 4.3, the slanted-vector-field construction, and Key Vanishing Lemma II hold. Let f be algebraically nondegenerate. The threshold and irreducible-component argument give a section ω_1 with $m<5t$ and residual irreducible divisor Z modulo Γ_2. If $f^*\omega_1\not\equiv 0$, Theorem 1.2 gives the result because $m/t<5$.

Assume $f^*\omega_1\equiv 0$. A nonconstant lifted curve is not contained in Γ_2, so analytic continuation gives $f_{[2]}(\mathbb{C})\subset Z$. If $t\geqslant m/5+7$, Proposition 5.3 either gives the constant-5 estimate or makes $f_{[1]}$ algebraically degenerate, in which case Theorem 1.1 is stronger. If $t/m<43/85$, then $\tau_1(t/m)>3/5$; choose $3/5<\tau<\tau_1(t/m)$. The restricted-section construction and Theorem 6.5 give coefficient $3/\tau<5$ when the canonical evaluation is nonzero, while Lemma 6.6 and Theorem 1.1 handle the zero-evaluation branch.

It remains only to see that one of these alternatives occurs. For $m=1,2$ this follows from Proposition 1.5. For $m\geqslant 23$ the two numerical regions overlap. For $3\leqslant m\leqslant 22$, the eighteen minimal uncovered pairs are exactly those excluded by Key Vanishing Lemma II; twist monotonicity excludes every larger twist with the same m. Finally, algebraic nondegeneracy implies $O(\log T_f(r)+\log r)=o(T_f(r))$ outside the usual exceptional set. This proves (1.7). □

7. Proof of the Key Vanishing Lemmas

7.1. **Basic Calculations.** Let $\mathscr{C} = \mathscr{C}_1 + \mathscr{C}_2 + \mathscr{C}_3$ be a divisor of three smooth conics in $\mathbb{P}^2$, defined by homogeneous quadratic polynomials:

$$\begin{aligned} A(Z) &= A_{200}Z_0^2 + A_{020}Z_1^2 + A_{002}Z_2^2 + A_{110}Z_0Z_1 + A_{101}Z_0Z_2 + A_{011}Z_1Z_2, \\ B(Z) &= B_{200}Z_0^2 + B_{020}Z_1^2 + B_{002}Z_2^2 + B_{110}Z_0Z_1 + B_{101}Z_0Z_2 + B_{011}Z_1Z_2, \\ C(Z) &= C_{200}Z_0^2 + C_{020}Z_1^2 + C_{002}Z_2^2 + C_{110}Z_0Z_1 + C_{101}Z_0Z_2 + C_{011}Z_1Z_2, \end{aligned}$$

where $Z = (Z_0, Z_1, Z_2)$ and all coefficients $A_\bullet, B_\bullet, C_\bullet \in \mathbb{C}$ are generic.

On the affine chart $U_0 = \{Z_0 \neq 0\}$ with coordinates $(x_1, x_2) = (Z_1/Z_0, Z_2/Z_0)$, the dehomogenized polynomials are:

$$\begin{aligned} a(x_1,x_2) &= A_{200} + A_{020}x_1^2 + A_{002}x_2^2 + A_{110}x_1 + A_{101}x_2 + A_{011}x_1x_2, \\ b(x_1,x_2) &= B_{200} + B_{020}x_1^2 + B_{002}x_2^2 + B_{110}x_1 + B_{101}x_2 + B_{011}x_1x_2, \\ c(x_1,x_2) &= C_{200} + C_{020}x_1^2 + C_{002}x_2^2 + C_{110}x_1 + C_{101}x_2 + C_{011}x_1x_2. \end{aligned}$$

Their partial derivatives are given by:

$$\begin{aligned} a_1 &:= \frac{\partial a}{\partial x_1} = 2A_{020}x_1 + A_{110} + A_{011}x_2, \quad a_2 := \frac{\partial a}{\partial x_2} = 2A_{002}x_2 + A_{101} + A_{011}x_1, \\ b_1 &:= \frac{\partial b}{\partial x_1} = 2B_{020}x_1 + B_{110} + B_{011}x_2, \quad b_2 := \frac{\partial b}{\partial x_2} = 2B_{002}x_2 + B_{101} + B_{011}x_1, \\ c_1 &:= \frac{\partial c}{\partial x_1} = 2C_{020}x_1 + C_{110} + C_{011}x_2, \quad c_2 := \frac{\partial c}{\partial x_2} = 2C_{002}x_2 + C_{101} + C_{011}x_1. \end{aligned}$$

The Jacobian determinant is the homogeneous cubic polynomial:

$$J(Z) := \begin{vmatrix} \frac{\partial}{\partial Z_0}A & \frac{\partial}{\partial Z_0}B & \frac{\partial}{\partial Z_0}C \\ \frac{\partial}{\partial Z_1}A & \frac{\partial}{\partial Z_1}B & \frac{\partial}{\partial Z_1}C \\ \frac{\partial}{\partial Z_2}A & \frac{\partial}{\partial Z_2}B & \frac{\partial}{\partial Z_2}C \end{vmatrix}.$$

Its vanishing locus $\{J = 0\}$ is the ramification divisor of the endomorphism:

$$\sigma\colon \mathbb{P}^2 \to \mathbb{P}^2, \quad [Z_0 : Z_1 : Z_2] \mapsto [A : B : C]. \tag{7.1}$$

The utility of such Jacobian curves has been well-documented in the literature, cf. e.g. [2, 13, 31, 32, 10].

Under generic coefficient assumptions, the Jacobian curve $\{J = 0\}$ satisfies the following transversality properties (cf., e.g., [32, 10]):

(**TP**) Transverse intersection with each conic $\mathscr{C}_i$ for $i = 1, 2, 3$.

For the Key Vanishing Lemma, we will work with parameters satisfying this transversality conditions.

We begin with the following two results, which follow from direct computation on the affine charts $\{Z_i \neq 0\} \subset \mathbb{P}^2$, $i = 0, 1, 2$.

Proposition 7.1. *Let $\mathscr{H} := \{Z_0 = 0\} + \{Z_1 = 0\} + \{Z_2 = 0\}$ be a divisor consisting of the three coordinate hyperplanes in $\mathbb{P}^2$. Then the logarithmic cotangent bundle $T^*_{\mathbb{P}^2}(\log \mathscr{H})$ is globally generated by the sections $(\log \frac{Z_0}{Z_2})'$ and $(\log \frac{Z_1}{Z_2})'$.* □

Proposition 7.2. *The logarithmic 2-jet differential bundle $E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{H})$ of weighted degree m is globally generated by the following sections:*

$$\left(\left(\log\frac{Z_0}{Z_2}\right)'\right)^k \left(\left(\log\frac{Z_1}{Z_2}\right)'\right)^{m-3j-k} \begin{vmatrix} \left(\log\frac{Z_0}{Z_2}\right)' & \left(\log\frac{Z_1}{Z_2}\right)' \\ \left(\log\frac{Z_0}{Z_2}\right)'' & \left(\log\frac{Z_1}{Z_2}\right)'' \end{vmatrix}^j \quad (0 \leqslant j \leqslant \lfloor m/3 \rfloor,\ 0 \leqslant k \leqslant m-3j). \tag{7.2}$$

□

The endomorphism σ defined in (7.1) is in fact a morphism $\sigma : (\mathbb{P}^2, \mathscr{C}) \to (\mathbb{P}^2, \mathscr{H})$ between two log pairs. The functoriality immediately yields the relations between logarithmic differential 1-forms

$$\left(\log\frac{A}{C}\right)' = \sigma^*\left(\log\frac{Z_0}{Z_2}\right)' \quad \text{and} \quad \left(\log\frac{B}{C}\right)' = \sigma^*\left(\log\frac{Z_1}{Z_2}\right)'.$$

Combining this with Proposition 7.1, we obtain:

Proposition 7.3. *On the Zariski-open set $\{J \neq 0\} \subset \mathbb{P}^2$, the logarithmic 1-forms $(\log \frac{A}{C})'$ and $(\log \frac{B}{C})'$ generate the sheaf $T^*_{\mathbb{P}^2}(\log \mathscr{C})$.*

We complement this conceptual proof with explicit computations. While the functorial approach is elegant, the detailed coordinate computations reveal specific algebraic structures that will play an essential role in our later analysis.

Proof. The sections $\left(\log \frac{A}{C}\right)'$ and $\left(\log \frac{B}{C}\right)'$ are clearly holomorphic on $\{J \neq 0\}$. We prove generation by working on affine charts.

On $U_0 = \{Z_0 \neq 0\}$ with coordinates (x_1, x_2), consider the transformation matrix between differentials:

$$\begin{pmatrix} \left(\log \frac{a}{c}\right)' \\ \left(\log \frac{b}{c}\right)' \end{pmatrix} = M \begin{pmatrix} x_1' \\ x_2' \end{pmatrix}, \quad M = \begin{pmatrix} \frac{\partial}{\partial x_1} \log \frac{a}{c} & \frac{\partial}{\partial x_2} \log \frac{a}{c} \\ \frac{\partial}{\partial x_1} \log \frac{b}{c} & \frac{\partial}{\partial x_2} \log \frac{b}{c} \end{pmatrix}.$$

A direct computation gives the determinant of M:

$$\det M = \frac{D}{abc}, \quad \text{where} \quad D = \begin{vmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \frac{J}{2Z_0^3}. \tag{7.3}$$

Thus the matrix M is invertible on the Zariski open set $U_0 \cap \{J \neq 0\} \cap \{ABC \neq 0\}$, with its inverse given explicitly by:

$$M^{-1} = \frac{1}{D} \begin{pmatrix} a(cb_2 - bc_2) & b(ac_2 - ca_2) \\ a(bc_1 - cb_1) & b(ca_1 - ac_1) \end{pmatrix}.$$

Hence inverting the relation (7.3) yields the explicit coordinate transformation:

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \frac{1}{D} \begin{pmatrix} a(cb_2 - bc_2) & b(ac_2 - ca_2) \\ a(bc_1 - cb_1) & b(ca_1 - ac_1) \end{pmatrix} \begin{pmatrix} \left(\log \frac{a}{c}\right)' \\ \left(\log \frac{b}{c}\right)' \end{pmatrix}. \tag{7.4}$$

This establishes that the logarithmic differential forms $\left(\log \frac{A}{C}\right)'$ and $\left(\log \frac{B}{C}\right)'$ generate the sheaf of logarithmic cotangent bundle $T^*_{\mathbb{P}^2}(\log \mathscr{C})$ over the Zariski-open set $\{J \neq 0\} \cap \{ABC \neq 0\}$, as the transformation matrix consists of regular functions on this domain.

Furthermore, the differential $(\log c)'$ can be expressed in terms of our generators:

$$\begin{aligned} (\log c)' = \frac{c'}{c} &= \begin{pmatrix} \frac{c_1}{c} & \frac{c_2}{c} \end{pmatrix} \begin{pmatrix} x_1' \\ x_2' \end{pmatrix} \\ \text{[use (7.4)]} \quad &= \frac{1}{D} \begin{pmatrix} a \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} & b \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} \end{pmatrix} \begin{pmatrix} \left(\log \frac{a}{c}\right)' \\ \left(\log \frac{b}{c}\right)' \end{pmatrix}. \end{aligned}$$

Thus the sections $\left(\log \frac{A}{C}\right)'$ and $\left(\log \frac{B}{C}\right)'$ continue to generate the logarithmic cotangent bundle $T^*_{\mathbb{P}^2}(\log \mathscr{C})$ in a neighborhood of $\{J \neq 0\} \cap \{C = 0\}$ inside $\{J \neq 0\}$, as the transformation matrix remains regular when approaching this locus. The analogous argument holds on charts $U_1 = \{Z_1 \neq 0\}$ and $U_2 = \{Z_2 \neq 0\}$, completing the proof. □

Using Proposition 7.2 and considering the morphism $\sigma : (\mathbb{P}^2, \mathscr{C}) \to (\mathbb{P}^2, \mathscr{H})$, by functoriality, we obtain:

Proposition 7.4. *The bundle $E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C})$ is freely generated over $\{J \neq 0\}$ by the logarithmic jet differentials:*

$$\left(\left(\log \frac{A}{C}\right)'\right)^k \left(\left(\log \frac{B}{C}\right)'\right)^{m-3j-k} \begin{vmatrix} \left(\log \frac{A}{C}\right)' & \left(\log \frac{B}{C}\right)' \\ \left(\log \frac{A}{C}\right)'' & \left(\log \frac{B}{C}\right)'' \end{vmatrix}^j \quad (0 \leqslant j \leqslant \lfloor m/3 \rfloor,\ 0 \leqslant k \leqslant m - 3j). \tag{7.5}$$

□

Let $P \in \Gamma\big(\{J \neq 0\}, E_{2,m}T^*\mathbb{P}^2(\log \mathscr{C}) \otimes \mathscr{O}(-t)\big)$. To check and verify the global extendability of P, it suffices to demonstrate that P admits a holomorphic extension across the Jacobian divisor $\{J = 0\}$. Note that the intersection $\{J = 0\} \cap \{Z_0 ABC = 0\}$ constitutes a Zariski-closed subset of codimension 2 in $\mathbb{P}^2$. By Hartog's extension theorem, it is thus sufficient to extend P to $\{J = 0\} \cap \{Z_0 ABC \neq 0\}$. This extension problem necessitates an analysis of the transition formula for $E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}(-t)$ from $\{J \neq 0\}$ to $\{Z_0 ABC \neq 0\}$.

Our carefully chosen basis (7.5) induces a highly homogeneous representation of the logarithmic jet differentials and keeps the affine transition problem finite.

The remaining argument treats all Wronskian layers simultaneously. This is essential for the strengthened list, whose largest target has $m = 21$; a small-m pole bound does not determine the higher layers that occur there.

7.2. **All-layer finite shape and exact C++ verification.** In what follows, we will abbreviate the first-order jet transfer formulas from standard jets (x_1', x_2') to logarithmic jets $((\log \frac{a}{c})', (\log \frac{b}{c})')$ as:

$$\begin{aligned} \left(\log \frac{a}{c}\right)' &= \alpha x_1' + \beta x_2', & \alpha &:= \frac{a_1 c - a c_1}{ac}, \quad \beta := \frac{a_2 c - a c_2}{ac}, \\ \left(\log \frac{b}{c}\right)' &= \gamma x_1' + \delta x_2', & \gamma &:= \frac{b_1 c - b c_1}{bc}, \quad \delta := \frac{b_2 c - b c_2}{bc}. \end{aligned}$$

Because logarithmic jet differentials are by definition assumed to exist and to be holomorphic outside the divisor $\mathscr{C} = \{ABC = 0\}$, the presence of a, b, c (within the affine chart $\mathbb{C}^2 \subset \mathbb{P}^2$) in denominators does not produce any singularity.

The only possible singularity—which clarifies why Proposition 7.4 holds only on the complement $\mathbb{P}^2 \setminus \{J = 0\}$—comes from inverting the transfer formulas from logarithmic jets back to standard jets:

$$\begin{aligned} x_1' &= \frac{\delta}{\alpha\delta - \beta\gamma}\left(\log \frac{a}{c}\right)' - \frac{\beta}{\alpha\delta - \beta\gamma}\left(\log \frac{b}{c}\right)', \\ x_2' &= -\frac{\gamma}{\alpha\delta - \beta\gamma}\left(\log \frac{a}{c}\right)' + \frac{\alpha}{\alpha\delta - \beta\gamma}\left(\log \frac{b}{c}\right)', \end{aligned} \tag{7.6}$$

and this requires the nonvanishing of the 2×2 determinant (see (7.3)):

$$\alpha\delta - \beta\gamma = \det \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \det M = \frac{D}{abc}, \quad \text{where} \quad D = \begin{vmatrix} a & b & c \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \frac{J}{2Z_0^3}. \tag{7.7}$$

By a direct computation in the affine chart $\{Z_0 \neq 0\}$, this determinant happens to be a nonzero multiple of the Jacobian determinant J.

The transfer of the logarithmic Wronskian,

$$\overline{W}''' := \begin{vmatrix} \left(\log \frac{A}{C}\right)' & \left(\log \frac{B}{C}\right)' \\ \left(\log \frac{A}{C}\right)'' & \left(\log \frac{B}{C}\right)'' \end{vmatrix} = \begin{vmatrix} \left(\log \frac{a}{c}\right)' & \left(\log \frac{b}{c}\right)' \\ \left(\log \frac{a}{c}\right)'' & \left(\log \frac{b}{c}\right)'' \end{vmatrix},$$

to standard affine jet coordinates is more delicate, and is captured in the following Key Observation.

Key Observation 7.8. *When expressed in standard affine jet coordinates $(x_1', x_2', x_1'', x_2'')$, the logarithmic Wronskian takes the form:*

$$\overline{W}''' = (\alpha\delta - \beta\gamma)\begin{vmatrix} x_1' & x_2' \\ x_1'' & x_2'' \end{vmatrix} + \kappa {x_1'}^3 + \lambda {x_1'}^2 x_2' + \mu x_1' {x_2'}^2 + \nu {x_2'}^3, \tag{7.9}$$

where the cubic terms in the first-order jets x_1', x_2' have coefficients κ, λ, μ, ν given in terms of the second-order derivatives

$$a_{11},\ a_{12},\ a_{22}, \qquad b_{11},\ b_{12},\ b_{22}, \qquad c_{11},\ c_{12},\ c_{22},$$

and the coefficients α, β, γ, δ as follows:

$$\begin{aligned}
\kappa &:= -\gamma\frac{a_{11}}{a} + \alpha\frac{b_{11}}{b} - (\alpha-\gamma)\frac{c_{11}}{c} + \alpha^2\gamma - \alpha\gamma^2,\\
\lambda &:= -\delta\frac{a_{11}}{a} - 2\gamma\frac{a_{12}}{a} + \beta\frac{b_{11}}{b} + 2\alpha\frac{b_{12}}{b} - (\beta-\delta)\frac{c_{11}}{c} - 2(\alpha-\gamma)\frac{c_{12}}{c}\\
&\quad + \alpha^2\delta + 2\alpha\beta\gamma - 2\alpha\gamma\delta - \beta\gamma^2,\\
\mu &:= -2\delta\frac{a_{12}}{a} - \gamma\frac{a_{22}}{a} + 2\beta\frac{b_{12}}{b} + \alpha\frac{b_{22}}{b} - 2(\beta-\delta)\frac{c_{12}}{c} - (\alpha-\gamma)\frac{c_{22}}{c}\\
&\quad + 2\alpha\beta\delta - \alpha\delta^2 + \beta^2\gamma - 2\beta\gamma\delta,\\
\nu &:= -\delta\frac{a_{22}}{a} + \beta\frac{b_{22}}{b} - (\beta-\delta)\frac{c_{22}}{c}\\
&\quad + \beta^2\delta - \beta\delta^2.
\end{aligned}$$

Proof. This follows by direct computation. □

According to Key Vanishing Lemma II, it is enough to consider $m \leqslant 21$. On the affine chart $(x_1, x_2) \in \mathbb{C}^2 \subset \mathbb{P}^2$, away from the Jacobian zero-locus

$$\{D = 0\} = \{J = 0\} \cap \{Z_0 \neq 0\},$$

the basis in Proposition 7.4 gives the uniform expansion

$$\mathsf{Jet}_{ABC} = \sum_{q=0}^{\lfloor m/3 \rfloor} \sum_{i+j=m-3q} P_{i,j}^{(q)} \Big(\big(\log \frac{a}{c}\big)' \Big)^i \Big(\big(\log \frac{b}{c}\big)' \Big)^j (\overline{W}''')^q. \tag{7.10}$$

Every coefficient $P_{i,j}^{(q)}$ is holomorphic away from $\{D = 0\}$.

The GAGA principle then ensures that every such coefficient is a rational expression whose denominator is an integral power of D. Our preliminary objective is to determine a uniform bound for that power in each Wronskian layer.

To this end, we consider the transfer formulas from logarithmic jets to standard jets. The first-order jet transfer formulas are already given in (7.6). Using (7.9), we have

$$W''' := \begin{vmatrix} x_1' & x_2' \\ x_1'' & x_2'' \end{vmatrix} = \frac{1}{\alpha\delta - \beta\gamma}\overline{W}''' - \frac{1}{\alpha\delta - \beta\gamma}\Big(\kappa {x_1'}^3 + \lambda {x_1'}^2 x_2' + \mu x_1' {x_2'}^2 + \nu {x_2'}^3 \Big),$$

where the cubic term $\kappa {x_1'}^3 + \lambda {x_1'}^2 x_2' + \mu x_1' {x_2'}^2 + \nu {x_2'}^3$ can be expanded using (7.6) as:

$$\begin{aligned}
&\begin{pmatrix} {x_1'}^3 & {x_1'}^2 x_2' & x_1' {x_2'}^2 & {x_2'}^3 \end{pmatrix} \begin{pmatrix} \kappa \\ \lambda \\ \mu \\ \nu \end{pmatrix}\\
&= \begin{pmatrix} \big(\big(\log \frac{a}{c}\big)'\big)^3 & \big(\big(\log \frac{a}{c}\big)'\big)^2 \big(\log \frac{b}{c}\big)' & \big(\log \frac{a}{c}\big)' \big(\big(\log \frac{b}{c}\big)'\big)^2 & \big(\big(\log \frac{b}{c}\big)'\big)^3 \end{pmatrix}\\
&\quad \cdot \frac{1}{(\alpha\delta - \beta\gamma)^3} \begin{pmatrix} \delta^3 & -\gamma\delta^2 & \gamma^2\delta & -\gamma^3 \\ -3\beta\delta^2 & \alpha\delta^2 + 2\beta\gamma\delta & -2\alpha\gamma\delta - \beta\gamma^2 & 3\alpha\gamma^2 \\ 3\beta^2\delta & -2\alpha\beta\delta - \beta^2\gamma & \alpha^2\delta + 2\alpha\beta\gamma & -3\alpha^2\gamma \\ -\beta^3 & \alpha\beta^2 & -\alpha^2\beta & \alpha^3 \end{pmatrix} \begin{pmatrix} \kappa \\ \lambda \\ \mu \\ \nu \end{pmatrix}.
\end{aligned}$$

Let us examine this more closely. Define

$$\begin{pmatrix} \delta^3 & -\gamma\delta^2 & \gamma^2\delta & -\gamma^3 \\ -3\beta\delta^2 & \alpha\delta^2 + 2\beta\gamma\delta & -2\alpha\gamma\delta - \beta\gamma^2 & 3\alpha\gamma^2 \\ 3\beta^2\delta & -2\alpha\beta\delta - \beta^2\gamma & \alpha^2\delta + 2\alpha\beta\gamma & -3\alpha^2\gamma \\ -\beta^3 & \alpha\beta^2 & -\alpha^2\beta & \alpha^3 \end{pmatrix} \begin{pmatrix} \kappa \\ \lambda \\ \mu \\ \nu \end{pmatrix} =: \begin{pmatrix} \tau_{3,0} \\ \tau_{2,1} \\ \tau_{1,2} \\ \tau_{0,3} \end{pmatrix}.$$

Miracle Factorizations 7.11. *The four expressions:*

$$\begin{aligned}
\tau_{3,0} &:= \delta^3\kappa - \gamma\delta^2\lambda + \gamma^2\delta\mu - \gamma^3\nu \\
&= (\alpha\delta - \beta\gamma)(\cdots), \\
\tau_{2,1} &:= -3\beta\delta^2\kappa + (\alpha\delta^2 + 2\beta\gamma\delta)\lambda + (-2\alpha\gamma\delta - \beta\gamma^2)\mu + 3\alpha\gamma^2\nu \\
&= (\alpha\delta - \beta\gamma)(\cdots), \\
\tau_{1,2} &:= 3\beta^2\delta\kappa + (-2\alpha\beta\delta - \beta^2\gamma)\lambda + (\alpha^2\delta + 2\alpha\beta\gamma)\mu - 3\alpha^2\gamma\nu \\
&= (\alpha\delta - \beta\gamma)(\cdots), \\
\tau_{0,3} &:= -\beta^3\kappa + \alpha\beta^2\lambda - \alpha^2\beta\mu + \alpha^3\nu \\
&= (\alpha\delta - \beta\gamma)(\cdots),
\end{aligned}$$

are all multiples of $\alpha\delta - \beta\gamma$.

Proof. Substitute the displayed formulas for $\kappa, \lambda, \mu, \nu$ and divide in the polynomial ring generated by $\alpha, \beta, \gamma, \delta$ and the normalized second derivatives. Each division has zero remainder. The four quotients are displayed and independently checked by exact symbolic expansion in Appendix A. □

Therefore:

$$\begin{pmatrix} \tau_{3,0} \\ \tau_{2,1} \\ \tau_{1,2} \\ \tau_{0,3} \end{pmatrix} =: (\alpha\delta - \beta\gamma) \begin{pmatrix} \theta_{3,0} \\ \theta_{2,1} \\ \theta_{1,2} \\ \theta_{0,3} \end{pmatrix},$$

and finally:

$$\begin{vmatrix} x_1' & x_2' \\ x_1'' & x_2'' \end{vmatrix} = \frac{1}{\alpha\delta - \beta\gamma} \begin{vmatrix} \left(\log\frac{a}{c}\right)' & \left(\log\frac{b}{c}\right)' \\ \left(\log\frac{a}{c}\right)'' & \left(\log\frac{b}{c}\right)'' \end{vmatrix} - \frac{1}{(\alpha\delta - \beta\gamma)^3} \sum_{i+j=3} \theta_{i,j} \left(\left(\log\frac{a}{c}\right)'\right)^i \left(\left(\log\frac{b}{c}\right)'\right)^j. \tag{7.12}$$

Recall that our main assumption is that, in the standard affine coordinates $(x_1, x_2) \in \mathbb{C}^2$, the logarithmic 2-jet differential Jet_{ABC} is holomorphic over $\mathbb{C}^2 \setminus \{abc = 0\}$. Therefore, it can be expressed as

$$\mathsf{Jet}_{ABC} = \sum_{q=0}^{\lfloor m/3 \rfloor} \sum_{r+s=m-3q} \overline{P}_{r,s}^{(q)} {x_1'}^r {x_2'}^s \left(W'''\right)^q, \tag{7.13}$$

where every $\overline{P}_{r,s}^{(q)}$ is holomorphic on $\mathbb{C}^2 \setminus \{abc = 0\}$. By the GAGA principle it is a rational function whose denominator may contain only a, b, and c.

We now compare (7.10) with (7.13). Substitution of (7.6) and (7.12) shows that the coefficients $\{P_{i,j}^{(q)}\}_{i+j=m-3q}$ can be written as rational expressions whose denominators consist of factors a, b, c (which are coprime to D) and D (see (7.7)). In Wronskian layer q, the maximal possible pole order with respect to D is $m - 2q$.

Indeed, an ordinary first derivative contributes at most one inverse power of D, while each occurrence of the lower cubic term in (7.12) trades three first-derivative weights for two additional inverse powers of D. For a monomial of total weight m in layer q, this gives precisely $D^{-(m-2q)}$. The argument is independent of q and therefore covers every $0 \leqslant q \leqslant \lfloor m/3 \rfloor$.

Consequently the invariant logarithmic 2-jet differential Jet_{ABC} in (7.10), of weighted order $m \leqslant 21$, has the following uniform all-layer form in homogeneous coordinates:

$$\mathsf{Jet}_{ABC} = \sum_{q=0}^{\lfloor m/3 \rfloor} \sum_{i+j=m-3q} \frac{P_{i,j}^{(q)}(Z_0, Z_1, Z_2)}{J(Z_0, Z_1, Z_2)^{m-2q}} \left(\left(\log\frac{A}{C}\right)'\right)^i \left(\left(\log\frac{B}{C}\right)'\right)^j \left(\overline{W}'''\right)^q, \tag{7.14}$$

where

$$n_q := m - 2q, \qquad P_{i,j}^{(q)} \in \mathbb{C}[Z_0, Z_1, Z_2]_{d_q}, \qquad d_q := 3n_q - t. \tag{7.15}$$

Indeed, homogeneity gives $d_q - 3(m - 2q) = -t$ in every layer. Thus (7.14) is not a truncation at a fixed Wronskian power: it is the complete finite shape, and it is precisely the shape enumerated by the C++ verifier. It represents a *meromorphic* (rational) section of

$$E_{2,m} T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-t).$$

For Jet_{ABC} to be a genuine *holomorphic* logarithmic jet differential, the coefficients of all polynomials $P_{i,j}^{(q)}$ must satisfy a finite homogeneous linear system. The system records that, after the complete coupled transition to ordinary affine jets, every coefficient has the divisibility required to cancel the appropriate power of the Jacobian J. Cancellations are taken only after summing the contributions of all Wronskian layers to the same ordinary-jet monomial; testing the layers separately would give only a diagnostic, not the desired kernel computation.

The number of homogeneous coefficient variables is

$$N(m,t)=\sum_{q=0}^{\lfloor m/3\rfloor}(m-3q+1)\binom{3(m-2q)-t+2}{2}, \tag{7.16}$$

where a summand is omitted when $3(m-2q)-t<0$. In particular, $N(13,9)=12{,}550$, whereas the largest strengthened target $(m,t)=(21,11)$ has $N(21,11)=81{,}840$ variables.

To streamline explicit computer-assisted computations, we adopt a strategic simplification by choosing specific coefficients $A_\bullet, B_\bullet, C_\bullet$ such that the three conics $\mathscr{C}$ are of Fermat type; for example:

$$A(Z)=2Z_0^2+Z_1^2+Z_2^2,\quad B(Z)=Z_0^2+2Z_1^2+Z_2^2,\quad C(Z)=Z_0^2+Z_1^2+2Z_2^2. \tag{7.17}$$

Each of the three conics in (7.17) is smooth. Any two meet transversally, and the three have no common point: subtracting their equations gives $Z_0^2=Z_1^2=Z_2^2$, after which any one equation forces all three coordinates to vanish. Hence their union is a simple normal crossing divisor. This choice also gives $J=32Z_0Z_1Z_2$, which makes the divisibility test particularly transparent.

Then we test extension on both $\{Z_0ABC\neq 0\}$ and $\{Z_2ABC\neq 0\}$. On $\mathbb{P}^2\setminus\mathscr{C}$, their union contains $\{J=0\}$ except for the single point $[0:1:0]$. That omitted set has codimension two, so Hartogs' theorem completes the extension there.

More precisely, we work in the two affine charts:

$$\{Z_0\neq 0\}\qquad\text{and}\qquad\{Z_2\neq 0\},$$

with the coordinate substitutions:

$$(Z_0,Z_1,Z_2)\longmapsto(1,x_1,x_2)\qquad\text{and}\qquad(Z_0,Z_1,Z_2)\longmapsto(y_0,y_1,1).$$

In these coordinates, J reads as $32x_1x_2$ or $32y_0y_1$. Thus the required cancellation is equivalent to divisibility by $(x_1x_2)^{m-2k}$ or $(y_0y_1)^{m-2k}$ in the coefficient of the ordinary Wronskian power W^k. These conditions form the finite homogeneous linear system computed below.

The exact matrix computed by the C++ verifier. For $\nu\in\{0,2\}$, let $U_\nu=\{Z_\nu\neq 0\}$ and write

$$(u_1,u_2)=\begin{cases}(x_1,x_2), & \nu=0,\\ (y_0,y_1), & \nu=2.\end{cases}$$

Let a,b,c denote the dehomogenizations of the homogeneous quadrics A,B,C on the current chart. We suppress the chart index below, and likewise use the same symbols J and $P_{i,j}^{(q)}$ for their dehomogenizations. Put

$$H:=abc,\qquad \widehat{L}_1:=b(ca'-ac'),\qquad \widehat{L}_2:=a(cb'-bc'),$$

and define

$$D:=\det\begin{pmatrix}a & b & c\\ \partial_{u_1}a & \partial_{u_1}b & \partial_{u_1}c\\ \partial_{u_2}a & \partial_{u_2}b & \partial_{u_2}c\end{pmatrix},\qquad W:=u_1'u_2''-u_1''u_2'.$$

Then

$$H\big(\log(a/c)\big)'=\widehat{L}_1,\qquad H\big(\log(b/c)\big)'=\widehat{L}_2.$$

Define the cubic first-jet polynomial R_3 by

$$\widehat{L}_1(\widehat{L}_2)'-\widehat{L}_2(\widehat{L}_1)'=HDW+R_3. \tag{7.18}$$

Equivalently,

$$H^3\overline{W}'''=H^2DW+HR_3. \tag{7.19}$$

The program constructs R_3 from (7.18); it does not enter a precomputed formula for it.

Fix a source layer q in (7.14), and put

$$n_q = m-2q, \qquad s_q = m-3q, \qquad i+j=s_q, \qquad \deg P_{i,j}^{(q)} = 3n_q - t.$$

After substituting (7.19), the contribution of $P_{i,j}^{(q)}$ to the coefficient of W^k, $0 \leqslant k \leqslant q$, is placed over the common denominator $H^{m-2k}J^{m-2k}$. Let $T_{i,j,q}$ denote the (i,j,q)-summand in (7.14). Using

$$\left(\log\frac{A}{C}\right)' = \frac{\widehat{L}_1}{H}, \qquad \left(\log\frac{B}{C}\right)' = \frac{\widehat{L}_2}{H}, \qquad \overline{W}''' = \frac{HDW+R_3}{H^2},$$

and $i+j=m-3q$, we obtain, for $0 \leqslant k \leqslant q$,

$$[W^k]T_{i,j,q} = \frac{\binom{q}{k}P_{i,j}^{(q)}\widehat{L}_1^i\widehat{L}_2^jD^k\left(HJ^2R_3\right)^{q-k}}{H^{m-2k}J^{m-2k}}.$$

Thus the polynomial in the numerator

$$\binom{q}{k}P_{i,j}^{(q)}\widehat{L}_1^i\widehat{L}_2^jD^k\left(HJ^2R_3\right)^{q-k} \tag{7.20}$$

is the contribution of the source term $T_{i,j,q}$ to the common numerator of the W^k-coefficient. For fixed k and a fixed velocity monomial, the program first sums (7.20) over *all* $q \geqslant k$ and only then tests divisibility by J^{m-2k}. This order matters: cancellation between different Wronskian layers is part of the extension problem. A layer-by-layer rank test would therefore compute a different, generally stronger system.

Each scalar coefficient of each homogeneous polynomial $P_{i,j}^{(q)}$ is a column. A row is indexed by a chart, the power k, a velocity monomial, and a base monomial that is not divisible by the required power of the monomial Jacobian. The verifier generates these sparse columns on demand and performs exact column elimination over $\mathbb{F}_5$; it never materializes a dense matrix.

Before a rank computation, the self-test verifies the determinant identity for D, checks (7.19) directly on nonsingular jets in both charts, and checks the expected variable counts. The transition matrix has integer entries. Hence full column rank after reduction modulo 5 exhibits an integer maximal minor that is nonzero modulo 5; that minor is nonzero over $\mathbb{Q}$, and the original matrix has zero kernel over $\mathbb{C}$.

All eighteen matrices in Key Vanishing Lemma II have full column rank. For the largest case $(m,t) = (21,11)$, the program processes 81,840 columns, finds rank 81,840, and returns nullity zero.[1]

Remark 7.5. For the explicit Fermat-type configuration (7.17), our exact finite computations give the vanishings for $(m,t) = (1,1),(2,1),(3,1),(4,1),(5,2)$. For $(m,t) = (5,1)$, by contrast, they give

$$\dim H^0\left(\mathbb{P}^2, E_{2,5}T^*_{\mathbb{P}^2}(\log\mathscr{C}) \otimes \mathscr{O}_{\mathbb{P}^2}(-1)\right) = 9.$$

Let $\eta_1,\ldots,\eta_9$ be a basis of this space. The vanishings in weighted degrees $1,2,3,4$ exclude a common invariant jet generator of smaller positive weight. The standard minimal-weight argument therefore shows that the 9-dimensional space contains two algebraically independent members, say η and θ. If either pullback is nonzero, Theorem 1.2 gives the desired estimate with coefficient $m/t = 5$. If both pullbacks vanish, the common zero locus of η and θ has proper image in the first Semple stage; hence $f_{[1]}$ is algebraically degenerate and Theorem 1.1 applies. Thus the nine sections give a Second Main Theorem with constant 5 for the Fermat-type configuration (7.17). This special conclusion is not needed in the proof of Theorem 1.6.

Declaration of AI Use

AI tools were used in the preparation of this article as follows. The proof of Proposition 1.5 was provided by ChatGPT (model: GPT-5.6 Sol). The authors have carefully checked the proof and verified its mathematical correctness.

The C++ code used for the computations in this article was produced by Codex by translating and adapting Maple code written by the authors. The authors have checked the resulting C++ code and verified the computational results used in the article.

AI tools were also used to assist with writing and to improve the clarity and language of this article. The authors have reviewed the resulting text and take full responsibility for the mathematical content, the code and computational results, and the final wording of this article.

[1] The complete C++ evidence archive—source, mathematical explanation, certificates, hashes, and full logs—is available from https://xiesongyan.github.io/.

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Appendix A. Explicit quotients in the four factorization identities

We complete the calculation in Miracle Factorizations 7.11. Set

$$\widehat{a}_{ij} := \frac{a_{ij}}{a}, \qquad \widehat{b}_{ij} := \frac{b_{ij}}{b}, \qquad \widehat{c}_{ij} := \frac{c_{ij}}{c}, \qquad \Delta := \alpha\delta - \beta\gamma.$$

After substituting the formulas for $\kappa, \lambda, \mu, \nu$, direct expansion gives

$$\tau_{3,0} = \Delta\theta_{3,0}, \qquad \tau_{2,1} = \Delta\theta_{2,1}, \qquad \tau_{1,2} = \Delta\theta_{1,2}, \qquad \tau_{0,3} = \Delta\theta_{0,3},$$

where

$$\begin{aligned}
\theta_{3,0} &= \delta^2(\widehat{b}_{11} - \widehat{c}_{11}) - 2\delta\gamma(\widehat{b}_{12} - \widehat{c}_{12}) + \gamma^2(\widehat{b}_{22} - \widehat{c}_{22}),\\
\theta_{2,1} &= -\delta^2\widehat{a}_{11} + 2\delta\gamma\widehat{a}_{12} - \gamma^2\widehat{a}_{22} - 2\beta\delta\widehat{b}_{11} + 2(\alpha\delta + \beta\gamma)\widehat{b}_{12} - 2\alpha\gamma\widehat{b}_{22}\\
&\quad + (2\beta\delta + \delta^2)\widehat{c}_{11} - 2(\alpha\delta + \beta\gamma + \delta\gamma)\widehat{c}_{12} + (2\alpha\gamma + \gamma^2)\widehat{c}_{22} + \Delta^2,\\
\theta_{1,2} &= 2\beta\delta\widehat{a}_{11} - 2(\alpha\delta + \beta\gamma)\widehat{a}_{12} + 2\alpha\gamma\widehat{a}_{22} + \beta^2\widehat{b}_{11} - 2\alpha\beta\widehat{b}_{12} + \alpha^2\widehat{b}_{22}\\
&\quad - (\beta^2 + 2\beta\delta)\widehat{c}_{11} + 2(\alpha\beta + \alpha\delta + \beta\gamma)\widehat{c}_{12} - (\alpha^2 + 2\alpha\gamma)\widehat{c}_{22} - \Delta^2,\\
\theta_{0,3} &= -\beta^2(\widehat{a}_{11} - \widehat{c}_{11}) + 2\alpha\beta(\widehat{a}_{12} - \widehat{c}_{12}) - \alpha^2(\widehat{a}_{22} - \widehat{c}_{22}).
\end{aligned}$$

For completeness, we also verified these identities by exact symbolic computation. The calculation was performed with SymPy 1.14.0 in the polynomial ring

$$\mathbb{Z}[\alpha, \beta, \gamma, \delta, \widehat{a}_{11}, \widehat{a}_{12}, \widehat{a}_{22}, \widehat{b}_{11}, \widehat{b}_{12}, \widehat{b}_{22}, \widehat{c}_{11}, \widehat{c}_{12}, \widehat{c}_{22}].$$

For each of the four expressions, exact division by Δ returned the quotient displayed above and remainder zero. A second expansion of $\tau_{i,j} - \Delta\theta_{i,j}$ returned the zero polynomial. Thus the factorization is an identity in the coefficient ring, not merely a numerical check at sampled values.

Appendix B. Proof of the One-Jet Vanishing Theorem

Proof of Proposition 1.5. The proof has three steps. First, two natural logarithmic forms identify E as an elementary transform of the trivial rank-two bundle along a smooth ramification cubic. A Chern-class computation then determines the transform and allows us to prove that E is nef by the curve criterion. Finally, an intersection calculation on $\mathbb{P}(E)$ excludes every negative twist.

Choose homogeneous quadratic equations

$$\mathscr{C}_1 = \{A = 0\}, \qquad \mathscr{C}_2 = \{B = 0\}, \qquad \mathscr{C}_3 = \{C = 0\}.$$

For a generic triple, A, B, C have no common zero and therefore define a morphism

$$\sigma\colon \mathbb{P}^2 \longrightarrow \mathbb{P}^2, \qquad [Z_0 : Z_1 : Z_2] \longmapsto [A : B : C]. \tag{B.1}$$

The morphism σ is proper. If one of its fibers were positive-dimensional, that fiber would contain a curve Γ_0. The restriction of $\sigma^*\mathscr{O}_{\mathbb{P}^2}(1)$ to Γ_0 would then be trivial, whereas

$$\sigma^*\mathscr{O}_{\mathbb{P}^2}(1) \simeq \mathscr{O}_{\mathbb{P}^2}(2)$$

has positive degree on every curve. Thus σ is quasi-finite and hence, being proper, finite. If $\mathscr{H} = \{Y_0Y_1Y_2 = 0\}$ is the coordinate triangle in the target, then $\mathscr{C} = \sigma^{-1}(\mathscr{H})$. Hence the logarithmic forms

$$\eta_1 = \mathrm{d}\log\frac{A}{C}, \qquad \eta_2 = \mathrm{d}\log\frac{B}{C}$$

induce a bundle morphism

$$\varphi\colon \mathscr{O}_{\mathbb{P}^2}^{\oplus 2} \longrightarrow E.$$

Let

$$J = \det\left(\frac{\partial(A,B,C)}{\partial(Z_0,Z_1,Z_2)}\right) \in H^0\big(\mathbb{P}^2, \mathscr{O}_{\mathbb{P}^2}(3)\big), \qquad R = \{J = 0\},$$

and set

$$\Omega = Z_0\,\mathrm{d}Z_1 \wedge \mathrm{d}Z_2 - Z_1\,\mathrm{d}Z_0 \wedge \mathrm{d}Z_2 + Z_2\,\mathrm{d}Z_0 \wedge \mathrm{d}Z_1.$$

A direct homogeneous computation gives

$$\eta_1 \wedge \eta_2 = \frac{J}{2}\frac{\Omega}{ABC}. \tag{B.2}$$

Thus R is the degeneracy divisor of φ. For a generic triple it is a smooth cubic; the relevant generic locus is shown to be nonempty in Remark B.1. Along R the determinant vanishes, so $\operatorname{rank}\varphi \leqslant 1$. If the rank were zero at some point, all four entries of a local matrix would belong to the maximal ideal there, and its determinant would vanish to order at least two. This contradicts the smoothness of R. Hence $\operatorname{rank}\varphi = 1$ everywhere on R. Consequently, with $i\colon R \hookrightarrow \mathbb{P}^2$, there is a line bundle L on R and an exact sequence

$$0 \longrightarrow \mathscr{O}_{\mathbb{P}^2}^{\oplus 2} \xrightarrow{\varphi} E \longrightarrow i_*L \longrightarrow 0. \tag{B.3}$$

Indeed, elementary row and column operations reduce a local matrix of φ along R to $\operatorname{diag}(1,u)$, where $u = 0$ is a local equation of R.

The residue sequence

$$0 \longrightarrow \Omega^1_{\mathbb{P}^2} \longrightarrow E \longrightarrow \bigoplus_{j=1}^{3} \mathscr{O}_{\mathscr{C}_j} \longrightarrow 0$$

gives

$$\operatorname{ch}(E) = 2 + 3H - \frac{9}{2}H^2, \qquad c_1(E) = 3H, \qquad c_2(E) = 9H^2. \tag{B.4}$$

Grothendieck–Riemann–Roch for the smooth cubic $i\colon R \hookrightarrow \mathbb{P}^2$ gives the explicit identity

$$\operatorname{ch}(i_*L) = i_*\left(\operatorname{ch}(L)\frac{\operatorname{td}(R)}{i^*\operatorname{td}(\mathbb{P}^2)}\right) = 3H + \left(\deg L - \frac{R^2}{2}\right)H^2 = 3H + \left(\deg L - \frac{9}{2}\right)H^2.$$

Comparison with (B.3) and (B.4) gives

$$\deg L = 0. \tag{B.5}$$

For completeness, the same Chern-character computation also gives

$$\operatorname{ch}(S^mE) = (m+1) + \frac{3m(m+1)}{2}H - \frac{9m(m+1)}{4}H^2.$$

Applying Hirzebruch–Riemann–Roch after twisting by $\mathscr{O}_{\mathbb{P}^2}(-t)$ proves formula (1.5).

We next prove that E is nef. By the curve criterion, it is enough to show that, for every morphism $\nu\colon B\to\mathbb{P}^2$ from a smooth projective curve, every quotient line bundle of ν^*E has nonnegative degree. We first treat curves contained in R. Restrict (B.3) to R. Since R is Cartier,

$$\operatorname{Tor}_1^{\mathbb{P}^2}(i_*L,\mathscr{O}_R)\simeq L\otimes\mathscr{O}_R(-R),$$

and tensoring gives

$$0\longrightarrow L(-R)\longrightarrow\mathscr{O}_R^{\oplus 2}\longrightarrow E|_R\longrightarrow L\longrightarrow 0. \tag{B.6}$$

The constant-rank-one property shows that the first arrow is a subbundle inclusion. If M denotes the image of $\mathscr{O}_R^{\oplus 2}\to E|_R$, then

$$\begin{aligned}0\longrightarrow L(-R)\longrightarrow\mathscr{O}_R^{\oplus 2}\longrightarrow M\longrightarrow 0,\\ 0\longrightarrow M\longrightarrow E|_R\longrightarrow L\longrightarrow 0.\end{aligned} \tag{B.7}$$

Both L and M are line bundles, and

$$\deg M=-\deg L(-R)=R^2=9. \tag{B.8}$$

It follows that $E|_R$ is nef. Indeed, for a quotient line bundle $E|_R\twoheadrightarrow Q$, either $M\to Q$ is nonzero, in which case $\deg Q\geqslant\deg M=9$, or the quotient factors through L, in which case $Q\simeq L$ and $\deg Q=0$.

Let now $\nu\colon B\to\mathbb{P}^2$ be a nonconstant morphism from a smooth projective curve. If $\nu(B)\not\subset R$, tensoring (B.3) with $\mathscr{O}_B$ may produce $\operatorname{Tor}_1^{\mathbb{P}^2}(i_*L,\mathscr{O}_B)$. This is a torsion sheaf, and its image in the torsion-free sheaf $\mathscr{O}_B^{\oplus 2}$ is zero. Hence $\nu^*\varphi$ is injective; it is generically an isomorphism and fits into

$$0\longrightarrow\mathscr{O}_B^{\oplus 2}\longrightarrow\nu^*E\longrightarrow T\longrightarrow 0, \tag{B.9}$$

where T is torsion. For every quotient line bundle $\nu^*E\twoheadrightarrow Q$, the composite $\mathscr{O}_B^{\oplus 2}\to Q$ is nonzero; otherwise the quotient would factor through T, whereas $\operatorname{Hom}(T,Q)=0$. Thus Q has a nonzero global section and $\deg Q\geqslant 0$. If $\nu(B)\subset R$, the same conclusion follows from the nefness of $E|_R$. Therefore E is nef.

Finally, use the quotient convention for

$$p\colon Y=\mathbb{P}(E)\longrightarrow\mathbb{P}^2,\qquad \xi=c_1(\mathscr{O}_Y(1)),\qquad h=p^*H.$$

Then $p_*\mathscr{O}_Y(m)=S^mE$ for $m\geqslant 0$, and the nefness of E is equivalent to that of ξ. With the quotient convention, the projective-bundle relation is

$$\xi^2-\xi p^*c_1(E)+p^*c_2(E)=0.$$

Together with (B.4), it gives the intersection numbers

$$\xi^3=c_1(E)^2-c_2(E)=0,\qquad \xi^2h=c_1(E)\cdot H=3. \tag{B.10}$$

By the projection formula, a nonzero section in (1.6) would define an effective divisor $\Gamma\sim m\xi-th$ on Y. Since ξ is nef, the intersection of the two nef classes ξ,ξ with the effective surface Γ is nonnegative; hence $\xi^2\cdot\Gamma\geqslant 0$. On the other hand,

$$\xi^2\cdot\Gamma=m\xi^3-t\xi^2h=-3t<0.$$

This contradiction proves the asserted vanishing. □

Remark B.1 (Nonemptiness of the genericity conditions). The conditions used in the proof are Zariski open and nonempty. For example, take

$$A=Z_0^2+Z_1Z_2,\qquad B=Z_1^2+Z_2Z_0,\qquad C=Z_2^2+Z_0Z_1.$$

The conics are smooth, have no common point, and meet pairwise transversally. Indeed, if one coordinate vanishes, the three equations force all three to vanish; if none vanishes, multiplying $A=B=C=0$ gives $(Z_0Z_1Z_2)^2=-(Z_0Z_1Z_2)^2$, again a contradiction. For instance, if $A=B=0$ and $\nabla A=\lambda\nabla B$, then

$$2Z_0=\lambda Z_2,\qquad Z_2=2\lambda Z_1,\qquad Z_1=\lambda Z_0.$$

If $Z_0=0$, all coordinates vanish; otherwise $\lambda^3=1$, and then $A=3Z_0^2\neq 0$, a contradiction. The other pairs follow cyclically. Moreover,

$$J=-2\big(Z_0^3+Z_1^3+Z_2^3-5Z_0Z_1Z_2\big),$$

which is a smooth Hesse cubic. Indeed, a common zero of its three partial derivatives would satisfy

$$3Z_0^2=5Z_1Z_2,\qquad 3Z_1^2=5Z_2Z_0,\qquad 3Z_2^2=5Z_0Z_1.$$

Multiplication gives $27(Z_0Z_1Z_2)^2 = 125(Z_0Z_1Z_2)^2$ when no coordinate vanishes, while if one coordinate vanishes, the three identities force all of them to vanish. Both alternatives are impossible in $\mathbb{P}^2$.

APPENDIX C. A COHOMOLOGICAL OBSERVATION

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Let $\mathscr{C} \subset \mathbb{P}^2$ be three generic smooth conics with simple normal crossings. Let

$$\begin{aligned} 0 \subset S^m T^*_{\mathbb{P}^2}(\log \mathscr{C}) &= F^0(E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C})) \\ &\subset F^1(E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C})) \subset \cdots \\ &\subset F^{\lfloor m/3 \rfloor}(E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C})) \\ &= E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \end{aligned}$$

be the ascending filtration of subsheaves that defines the grading (2.5). We set $F^k(E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C})) = 0$ for $k \leqslant -1$. From (2.5) we obtain:

Proposition C.1. *For every $0 \leqslant k \leqslant m/3$, the following sequence is exact:*

$$\begin{aligned} 0 &\longrightarrow F^{k-1}(E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C})) \longrightarrow F^k(E_{2,m}T^*_{\mathbb{P}^2}(\log \mathscr{C})) \\ &\longrightarrow S^{m-3k}T^*_{\mathbb{P}^2}(\log \mathscr{C}) \otimes \overline{K}^{\otimes k}_{\mathbb{P}^2} \longrightarrow 0. \end{aligned}$$

For convenience, from now on we write $\overline{T}^*_{\mathbb{P}^2} \coloneqq T^*_{\mathbb{P}^2}(\log \mathscr{C})$ and $\mathscr{O}(1) \coloneqq \mathscr{O}_{\mathbb{P}^2}(1)$.

Proposition C.2. *For integer $q \geqslant 0$ and $r \in \{1,2\}$, the following vanishing holds:*

$$H^0\big(\mathbb{P}^2, E_{2,3q+r}\overline{T}^*_{\mathbb{P}^2} \otimes \mathscr{O}_{\mathbb{P}^2}(-(3q+r))\big) = 0,$$

provided that $H^0\big(\mathbb{P}^2, S^{3\ell+r}\overline{T}^*_{\mathbb{P}^2} \otimes \mathscr{O}_{\mathbb{P}^2}(-(3\ell+r))\big) = 0$ for all $\ell = 0,1,\ldots,q$.

Proof. By Proposition C.1, for every integer $k = 0,1,\ldots,q$, we have a short exact sequence:

$$\begin{aligned} 0 &\longrightarrow F^{k-1}(E_{2,3q+r}\overline{T}^*_{\mathbb{P}^2}) \otimes \mathscr{O}(-(3q+r)) \\ &\longrightarrow F^k(E_{2,3q+r}\overline{T}^*_{\mathbb{P}^2}) \otimes \mathscr{O}(-(3q+r)) \\ &\longrightarrow S^{3q+r-3k}\overline{T}^*_{\mathbb{P}^2} \otimes \overline{K}^{\otimes k}_{\mathbb{P}^2} \otimes \mathscr{O}(-(3q+r)) \longrightarrow 0. \end{aligned}$$

Note that the last term $S^{3q+r-3k}\overline{T}^*_{\mathbb{P}^2} \otimes \overline{K}^{\otimes k}_{\mathbb{P}^2} \otimes \mathscr{O}(-(3q+r)) \cong S^{3(q-k)+r}\overline{T}^*_{\mathbb{P}^2} \otimes \mathscr{O}(-(3(q-k)+r))$ has no global section by our assumption. Considering the induced long exact sequence in cohomology, we obtain

$$\begin{aligned} &H^0\Big(\mathbb{P}^2, F^k(E_{2,3q+r}\overline{T}^*_{\mathbb{P}^2}) \otimes \mathscr{O}(-(3q+r))\Big) \\ &\quad\cong H^0\Big(\mathbb{P}^2, F^{k-1}(E_{2,3q+r}\overline{T}^*_{\mathbb{P}^2}) \otimes \mathscr{O}(-(3q+r))\Big). \end{aligned}$$

Iterating backwards for $k = q, q-1, \ldots, 1$ gives the desired vanishing, since by assumption

$$H^0\big(\mathbb{P}^2, S^{3q+r}\overline{T}^*_{\mathbb{P}^2} \otimes \mathscr{O}(-(3q+r))\big) = 0.$$

□

Remark C.3. For $r = 0$ the argument fails: when $q = k$, the term

$$S^{3q+r-3k}\overline{T}^*_{\mathbb{P}^2} \otimes \overline{K}^{\otimes k}_{\mathbb{P}^2} \otimes \mathscr{O}(-(3q+r)) \cong \mathscr{O}_{\mathbb{P}^2}$$

has one-dimensional global sections.

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